

Facility Location Problems

Instructor: Sairam Sriraman

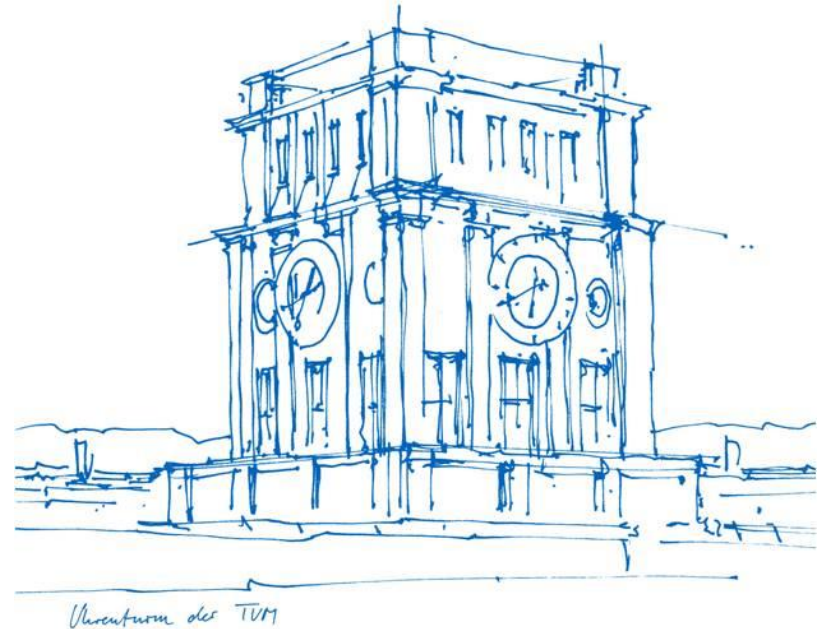
Examiner: Prof. Dr. David Wuttke

Technische Universität München

School of Management

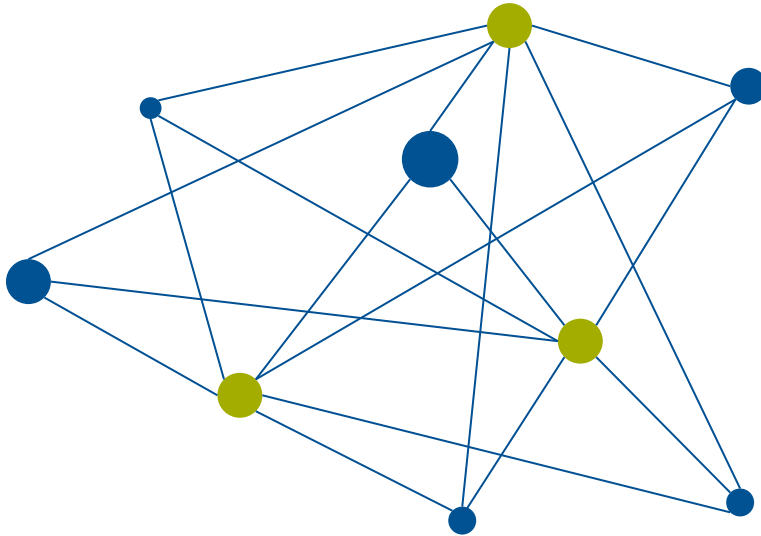
Campus Heilbronn

Heilbronn, 2026



Plant Location Model (Problem) and Linear Programming

How should you serve your customers?



- Plant location
- Customer location

- Each plant has a fixed capacity
- Each customer has fixed demand
- Each route has specific costs

Model

Number of plant facilities	m
Number of customer locations	n
Capacity of production facility i	K_i
Demand at customer location j	D_j
Costs to produce and ship from i to j	c_{ij}
Amount shipped from i to j	x_{ij}

$$\min \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

subject to:

$$\sum_{i=1}^m x_{ij} \geq D_j \quad j = 1, \dots, n$$

$$\sum_{j=1}^n x_{ij} \leq K_i \quad i = 1, \dots, m$$

$$x_{ij} \geq 0 \quad \begin{matrix} i = 1, \dots, m; \\ j = 1, \dots, n \end{matrix}$$

Graphical Solution

Graphical Solution – one Example

Number of plants facilities	$m = 2$
Number of customer locations	$n = 1$
Capacity of production facility i	$K_1 = 100, K_2 = 50$
Demand at customer location j	$D_1 = 30$
Costs to produce and ship from i to j	$c_{11} = 80, c_{21} = 120$
Amount shipped from i to j	$x_{11} = ?, x_{21} = ?$

$$\min \quad 80x_{11} + 120x_{21}$$

subject to:

$$x_{11} + x_{21} \geq 30$$

$$x_{11} \leq 100$$

$$x_{21} \leq 50$$

$$x_{11} \geq 0$$

$$x_{12} \geq 0$$

First Derivative, Second Derivative, What's the Matter?

$$\min \quad 80x_{11} + 120x_{21}$$

subject to:

$$x_{11} + x_{21} \geq 30$$

$$x_{11} \leq 100$$

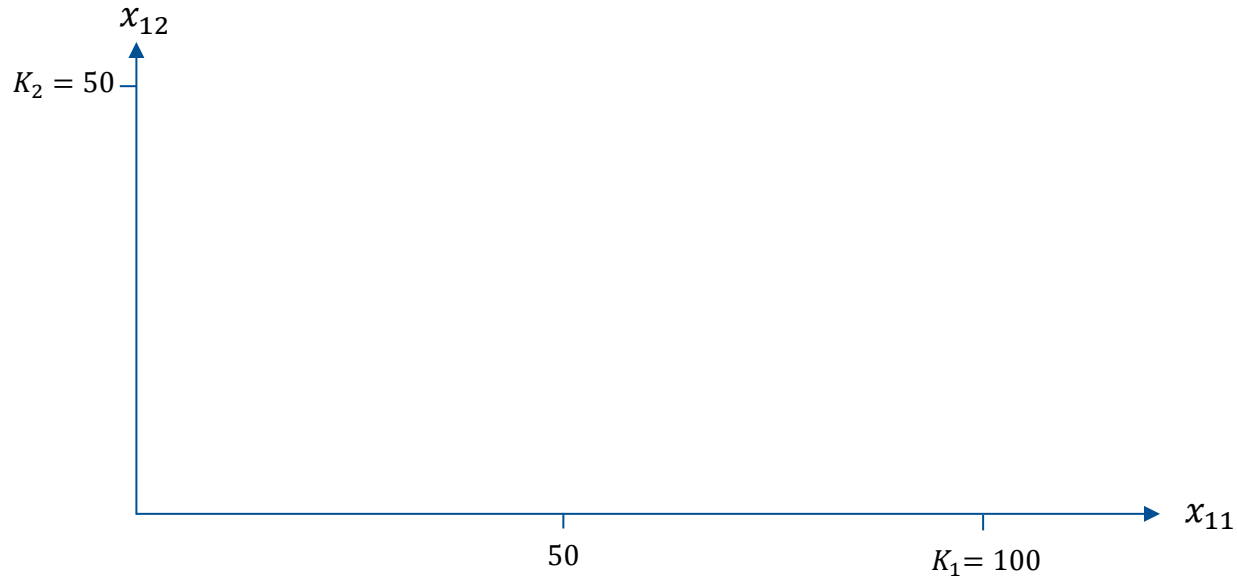
$$x_{21} \leq 50$$

$$x_{11} \geq 0$$

$$x_{12} \geq 0$$

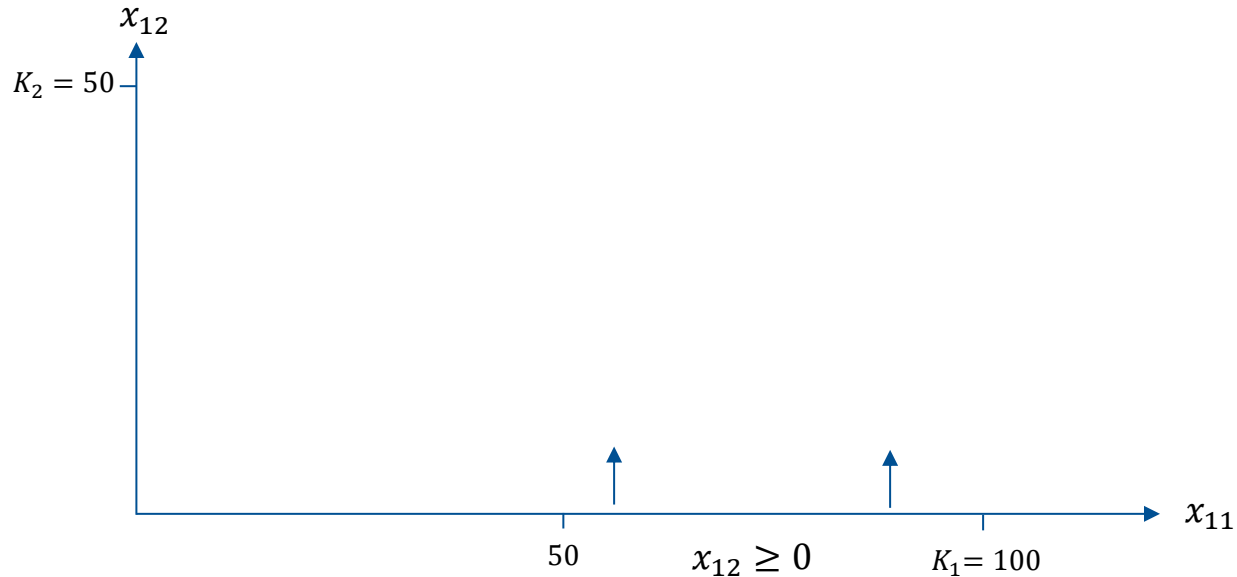
Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



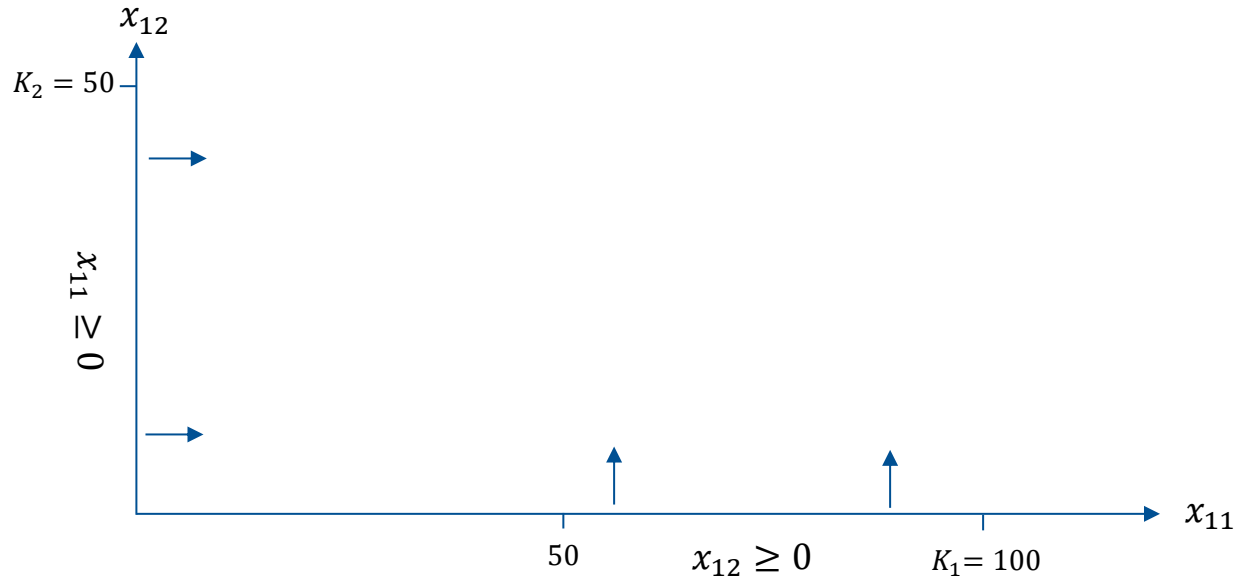
Graphical Solution

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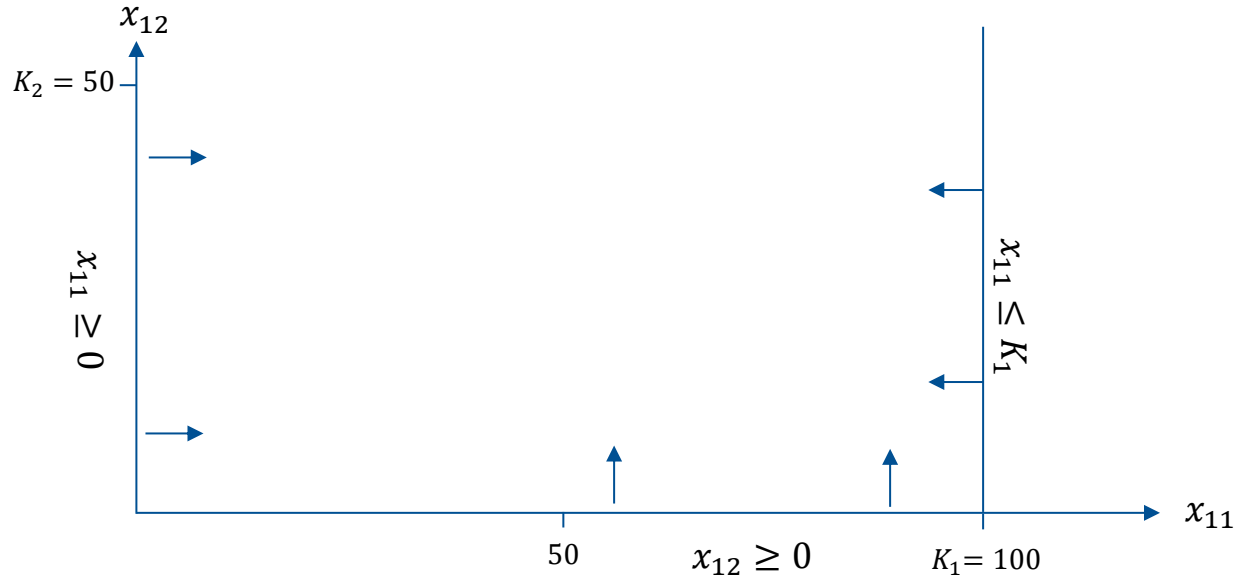
Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



Graphical Solution

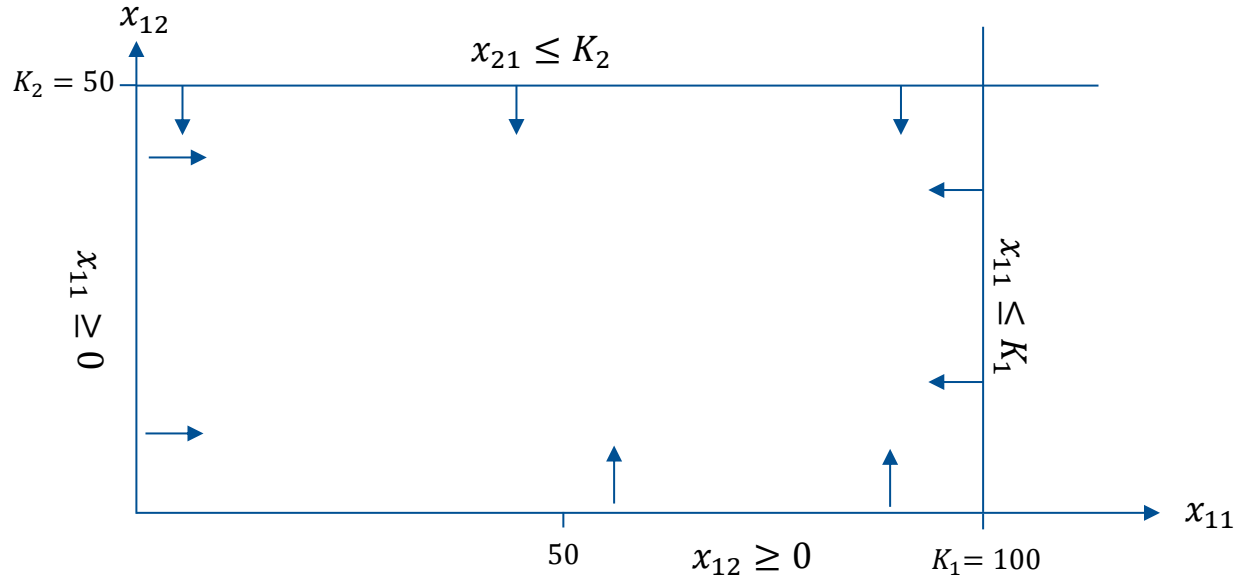
$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\sum_{j=1}^n x_{ij} \leq K_i \quad i = 1, \dots, m$$

Graphical Solution

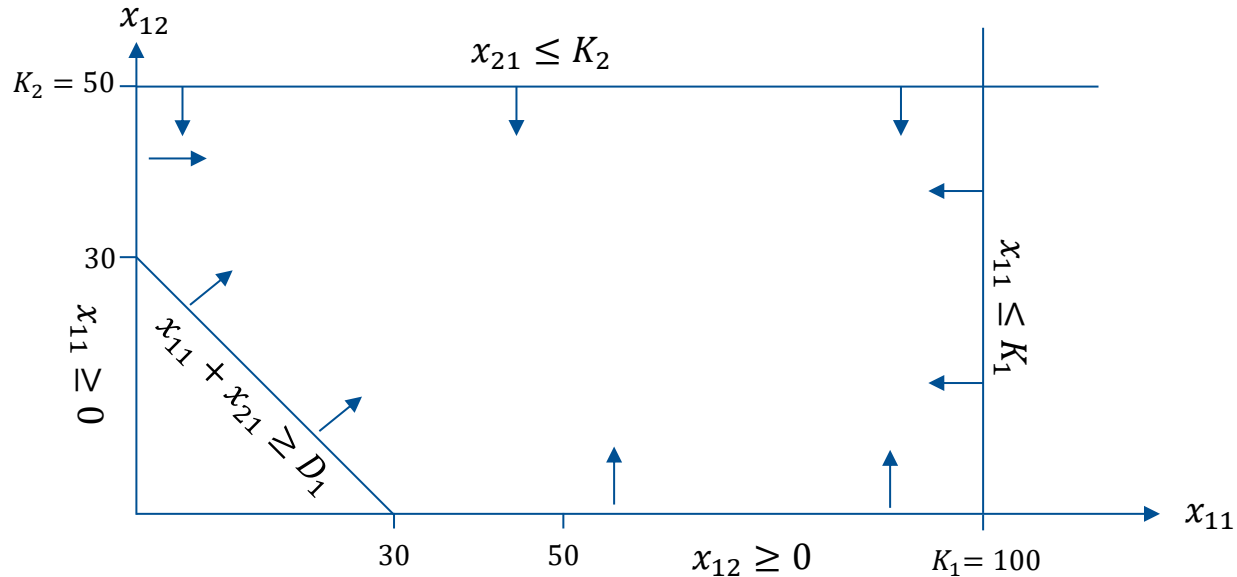
$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



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Graphical Solution

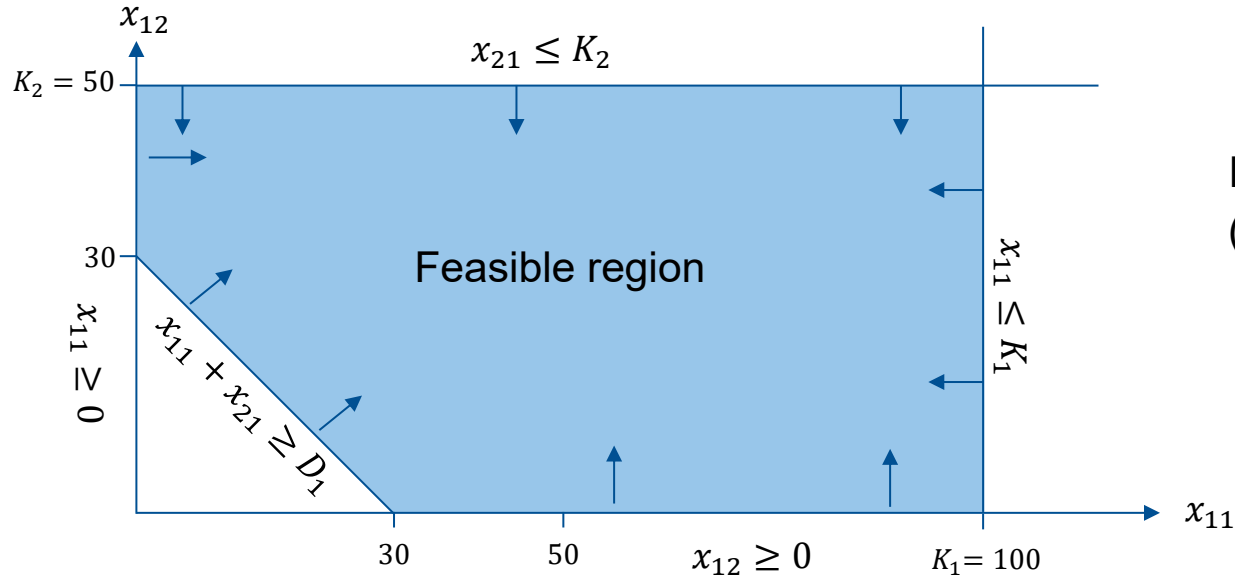
$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\sum_{i=1}^m x_{ij} \geq D_j \quad j = 1, \dots, n$$

Graphical Solution

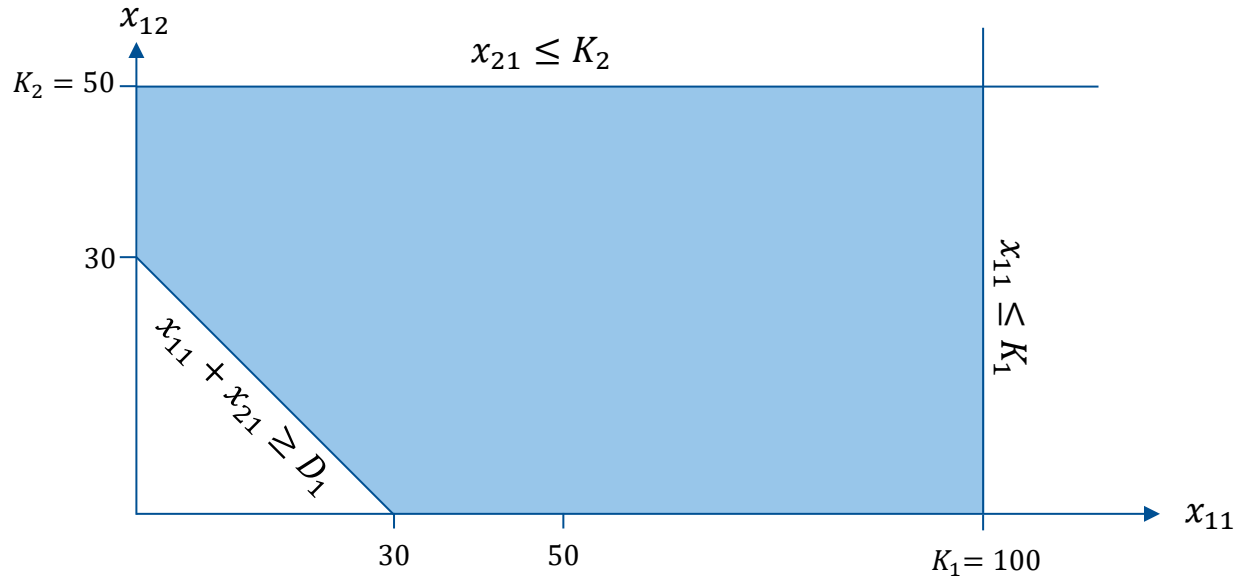
$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



Infeasible solutions
(entire white area)

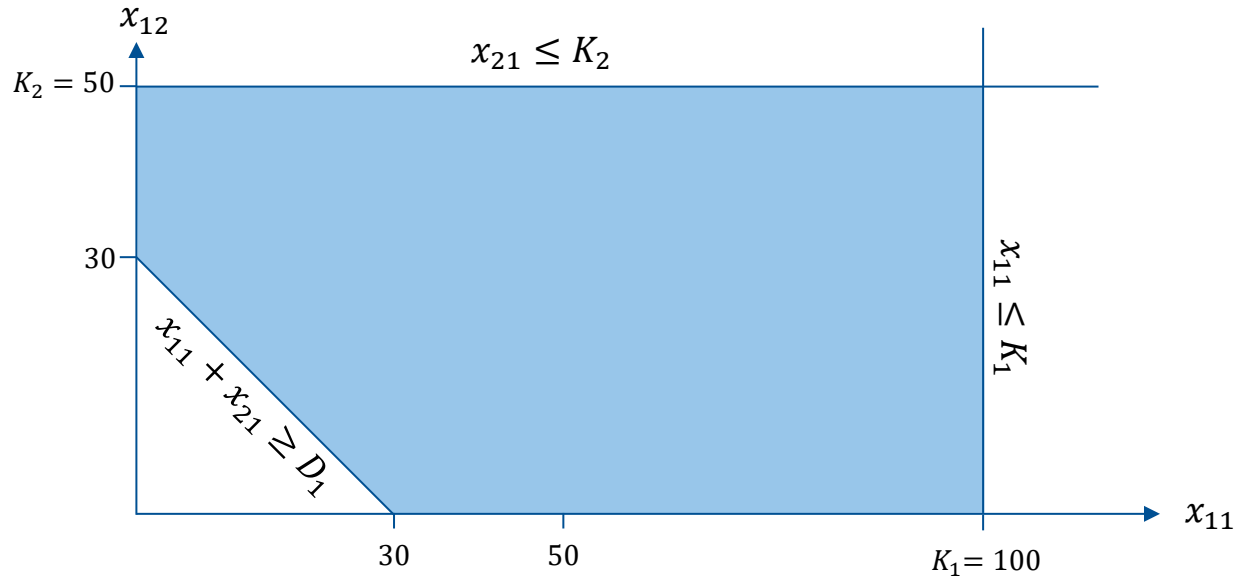
Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$

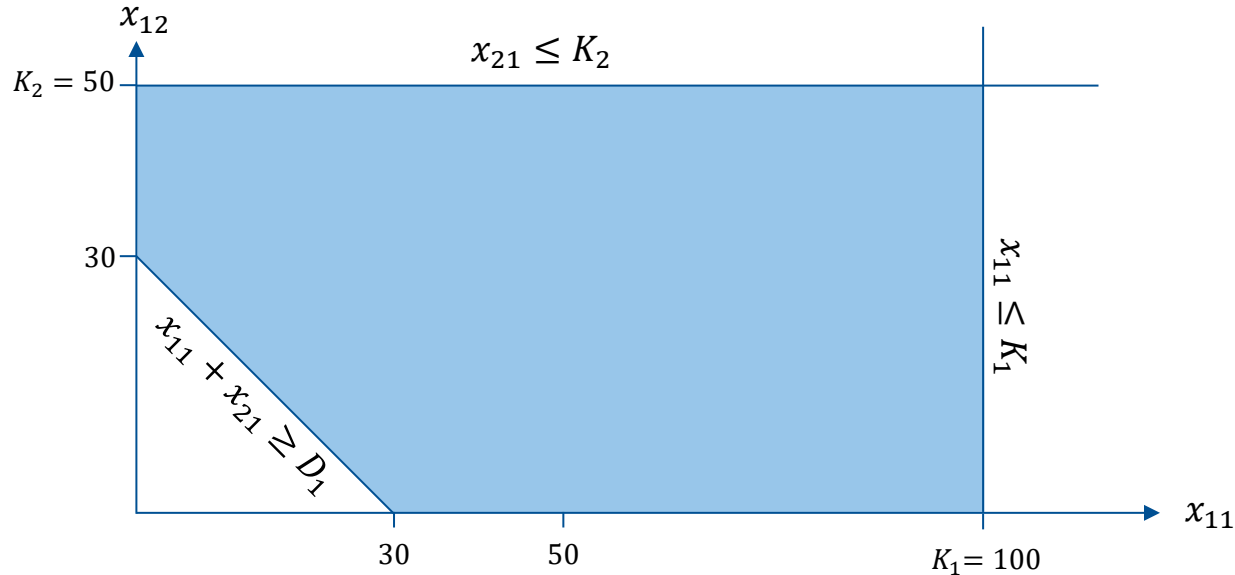


$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

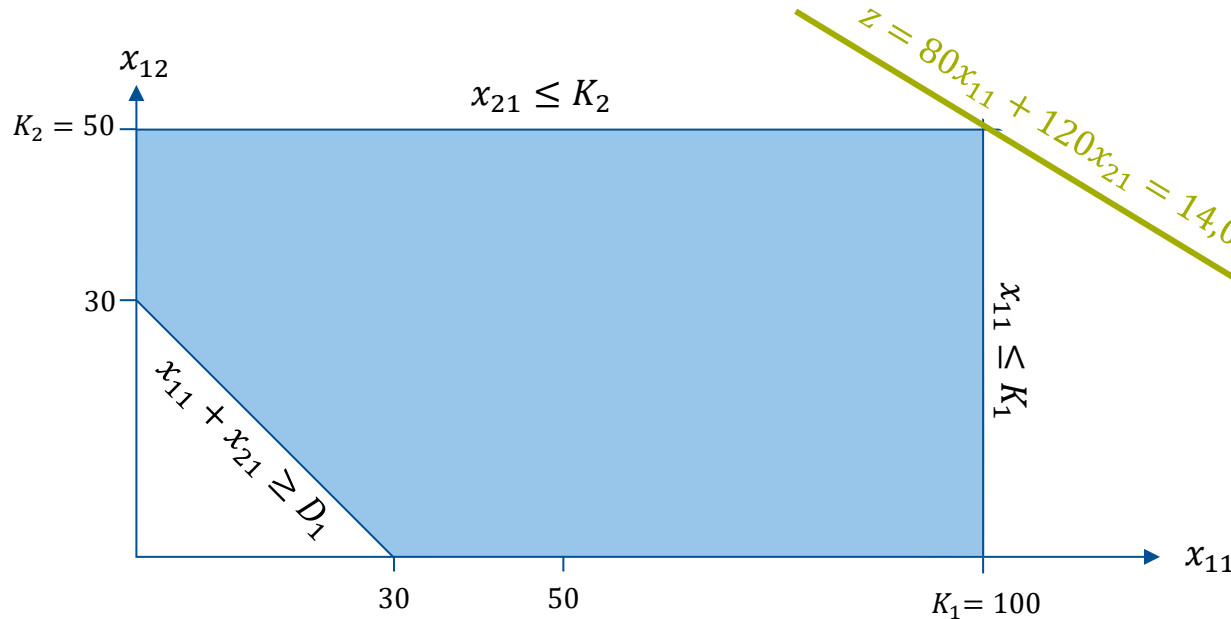
e.g.,

$$z = 80 \cdot 100 + 120 \cdot 50$$

$$z = 14,000$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

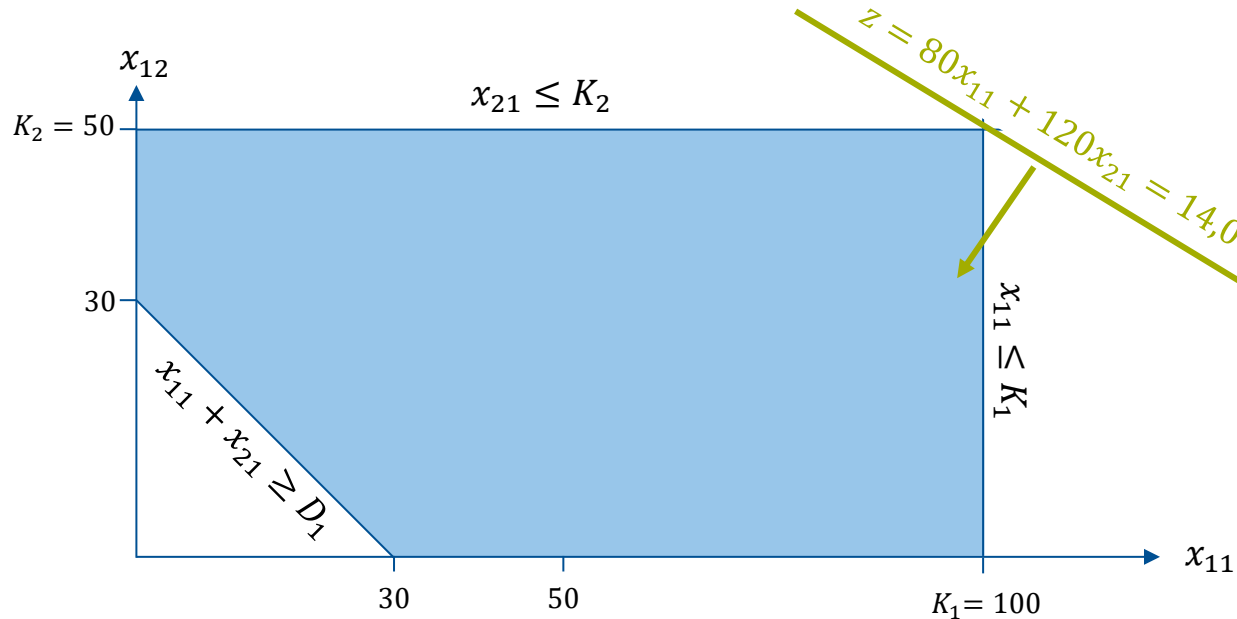
e.g.,

$$z = 80 \cdot 100 + 120 \cdot 50$$

$$z = 14,000$$

Graphical Solution

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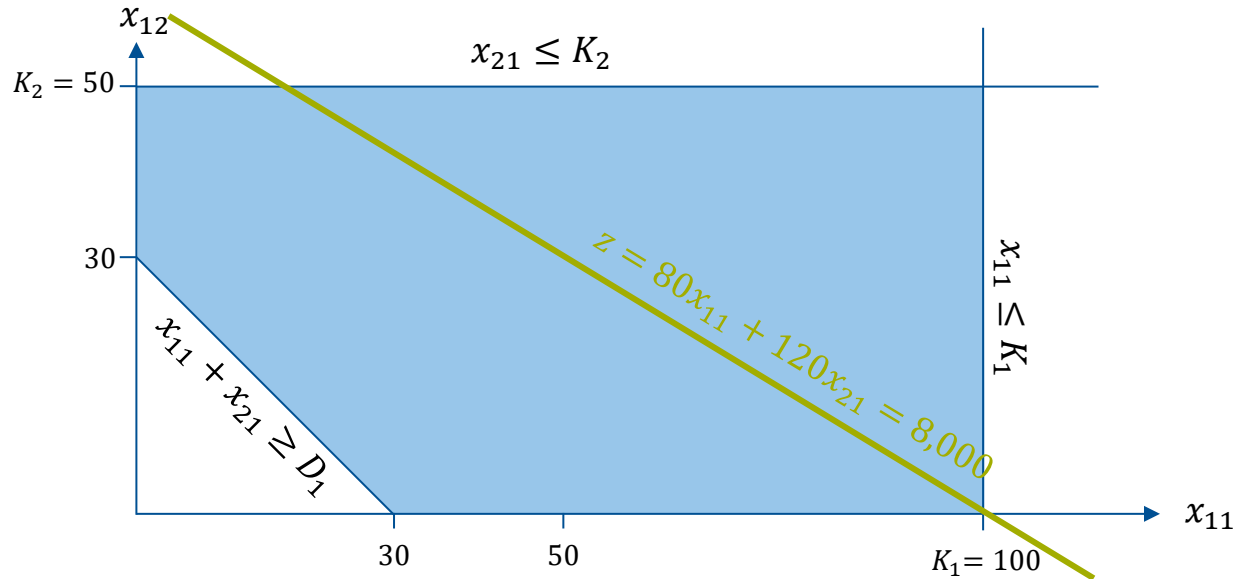
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Graphical Solution

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$$z = 80x_{11} + 120x_{21}$$

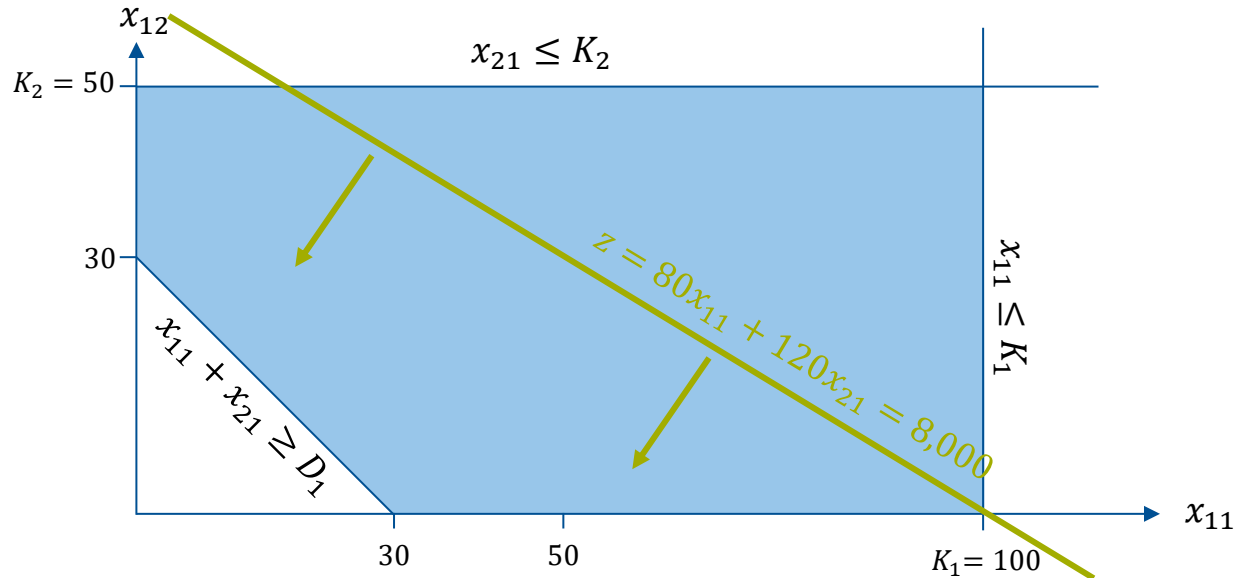
e.g.,

$$z = 80 \cdot 100 + 120 \cdot 0$$

$$z = 8,000$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

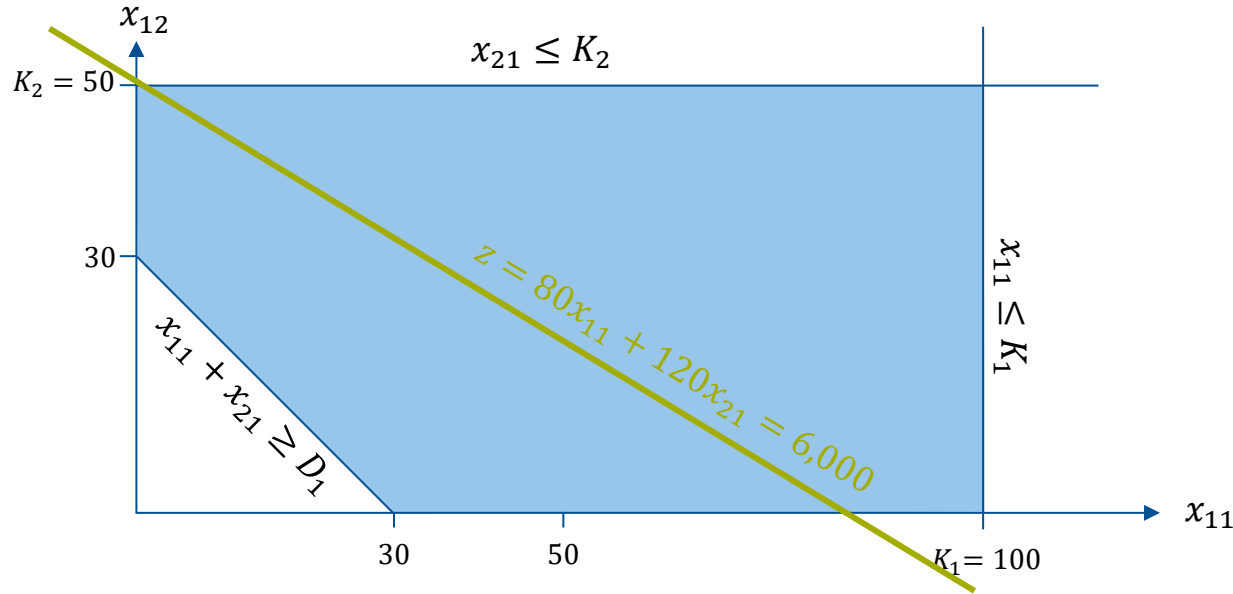
e.g.,

$$z = 80 \cdot 100 + 120 \cdot 0$$

$$z = 8,000$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

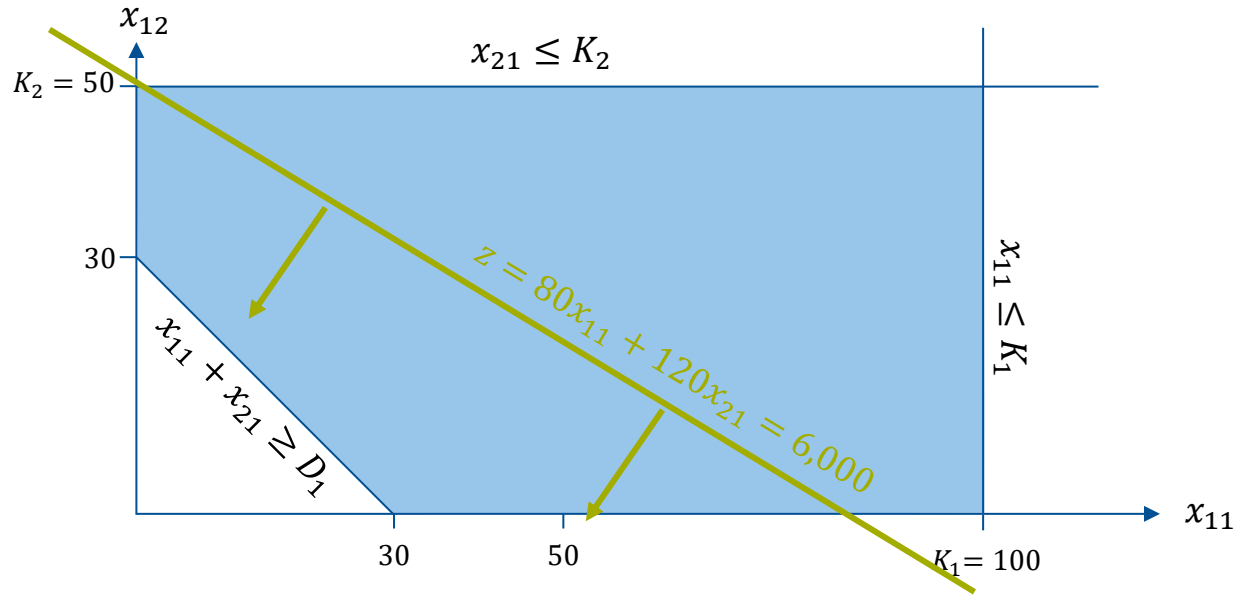
e.g.,

$$z = 80 \cdot 0 + 120 \cdot 50$$

$$z = 6,000$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

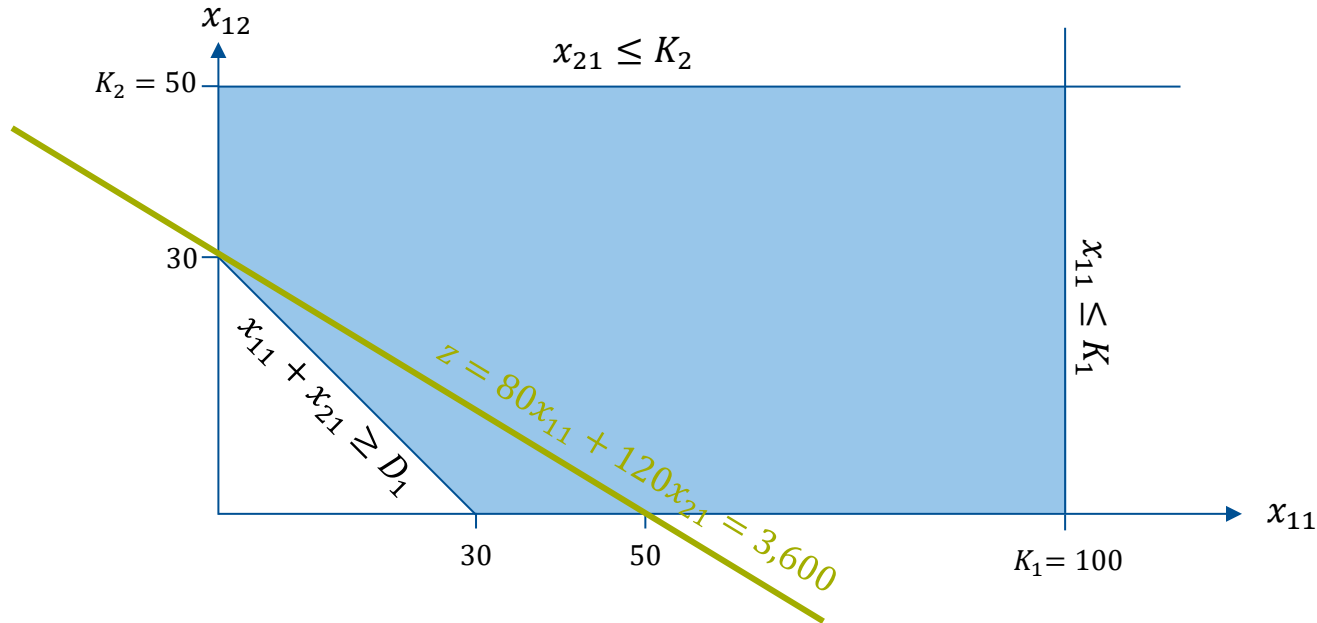
e.g.,

$$z = 80 \cdot 0 + 120 \cdot 50$$

$$z = 6,000$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

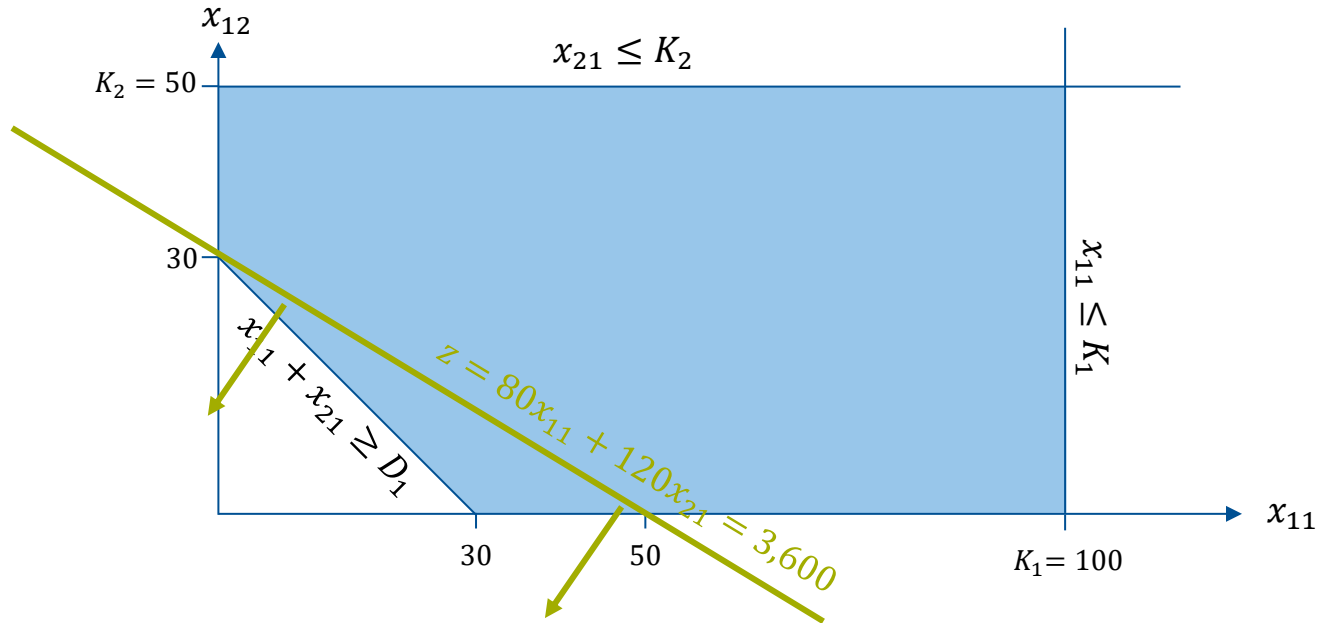
e.g.,

$$z = 80 \cdot 0 + 120 \cdot 30$$

$$z = 3,600$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

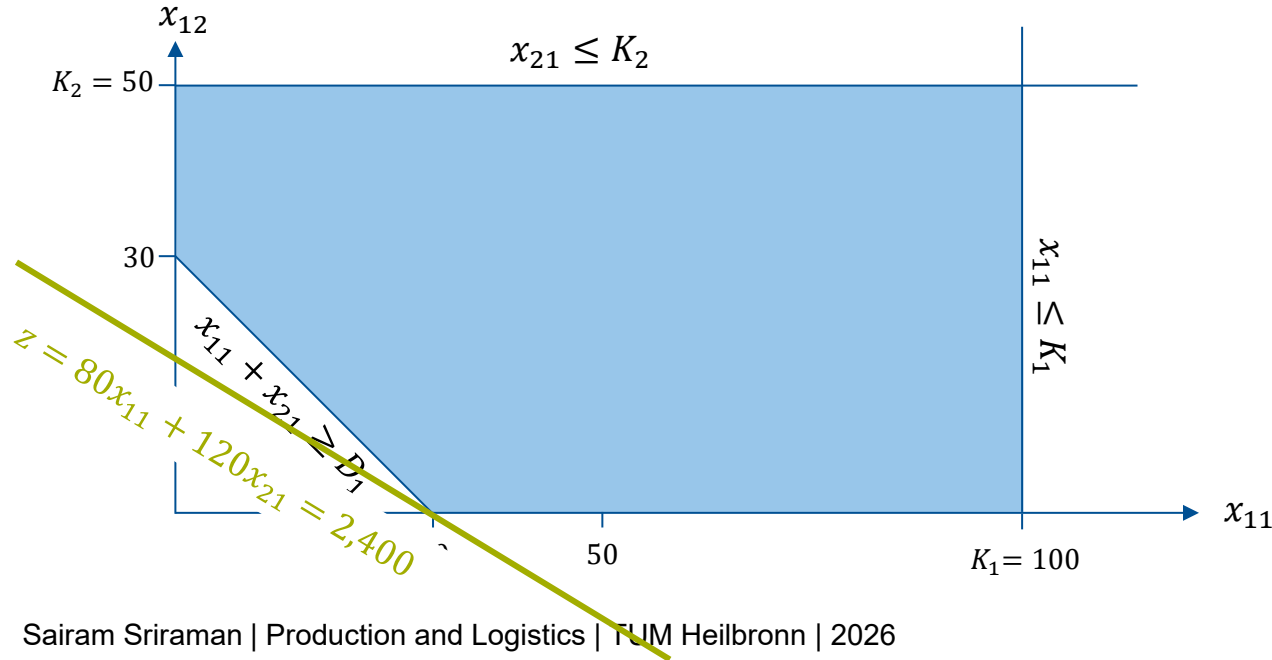
e.g.,

$$z = 80 \cdot 0 + 120 \cdot 30$$

$$z = 3,600$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



$$\min z := \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

$$z = 80x_{11} + 120x_{21}$$

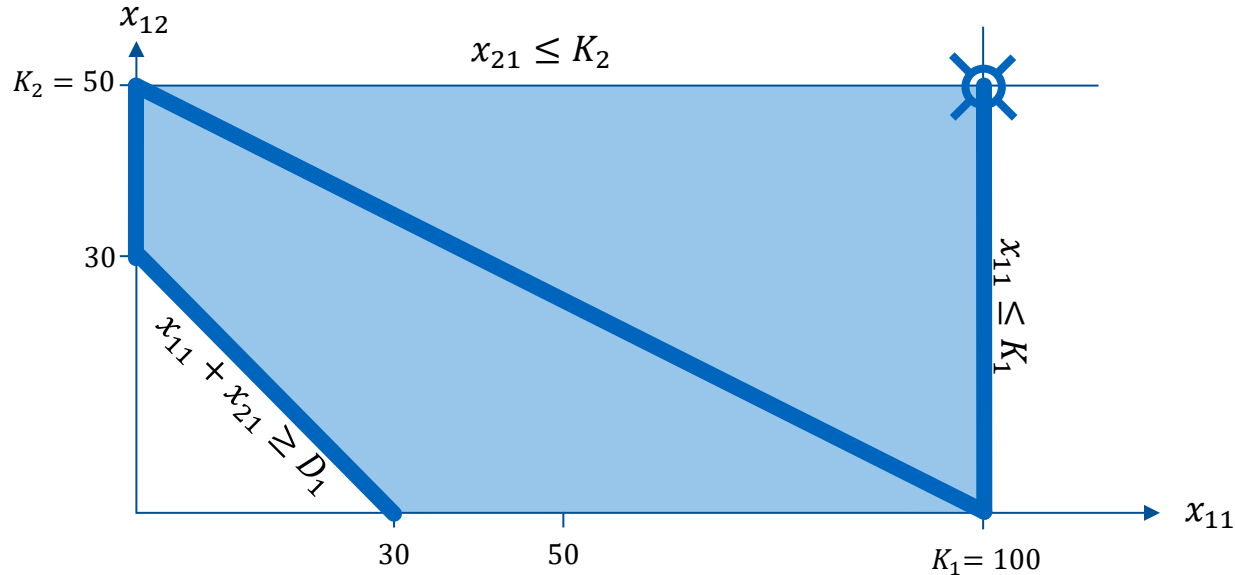
e.g.,

$$z = 80 \cdot 30 + 120 \cdot 0$$

$$z = 2,400$$

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



Limits to the (Graphical) Solution and Outlook

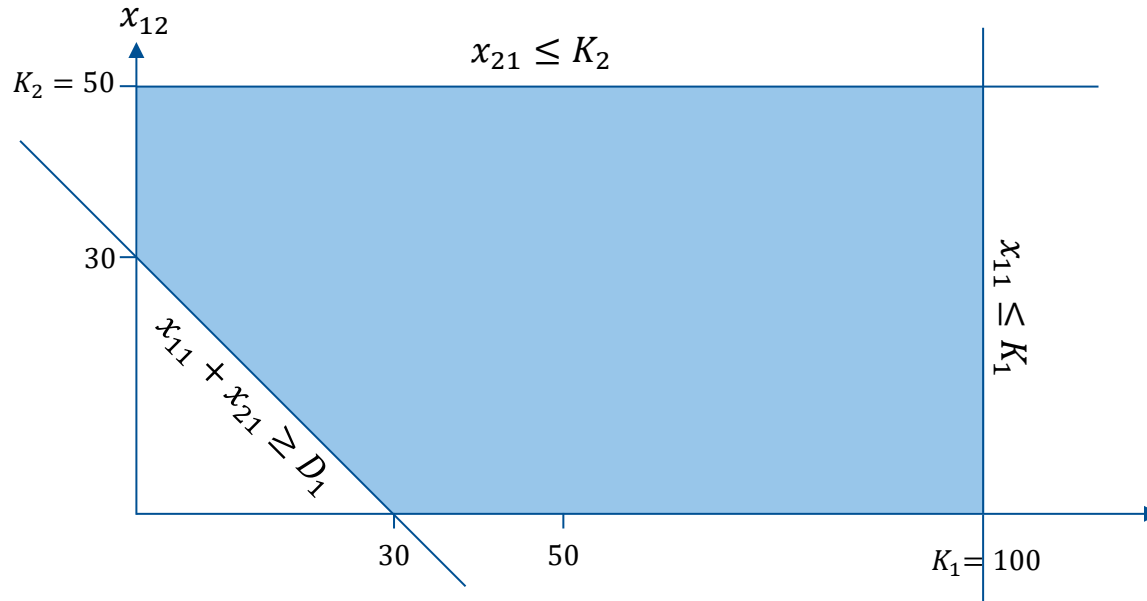
Graphical Solution?

Only works for 2 or maybe 3 variables.

Could we still implement the idea? That is, for higher dimensional problems, could we calculate all corners, then calculate for each of them the objective function and pick the best corner?

Graphical Solution?

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



Consider a variables with b linear equations* and $a > b$ then the **maximum** number of corners is

$$C_b^a = \frac{a!}{b!(a-b)!}.$$

For instance, here $a = 5$, $b = 3$ and $C_3^5 = 10$.

Here, there are indeed only 8 corners due to parallel constraints.

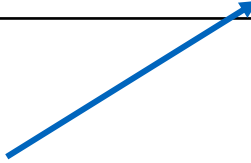
*) Note: above we consider linear inequalities. Each inequality can be converted into a linear equation by adding a slack variable per inequality, as we will do next session.

Graphical Solution?

Number of variables (a)	Number of linear equations (b)	C_b^a
5	2	10
20	10	184,756
49	6	13,983,816
82	41	$4.25 \cdot 10^{23}$

Graphical Solution?

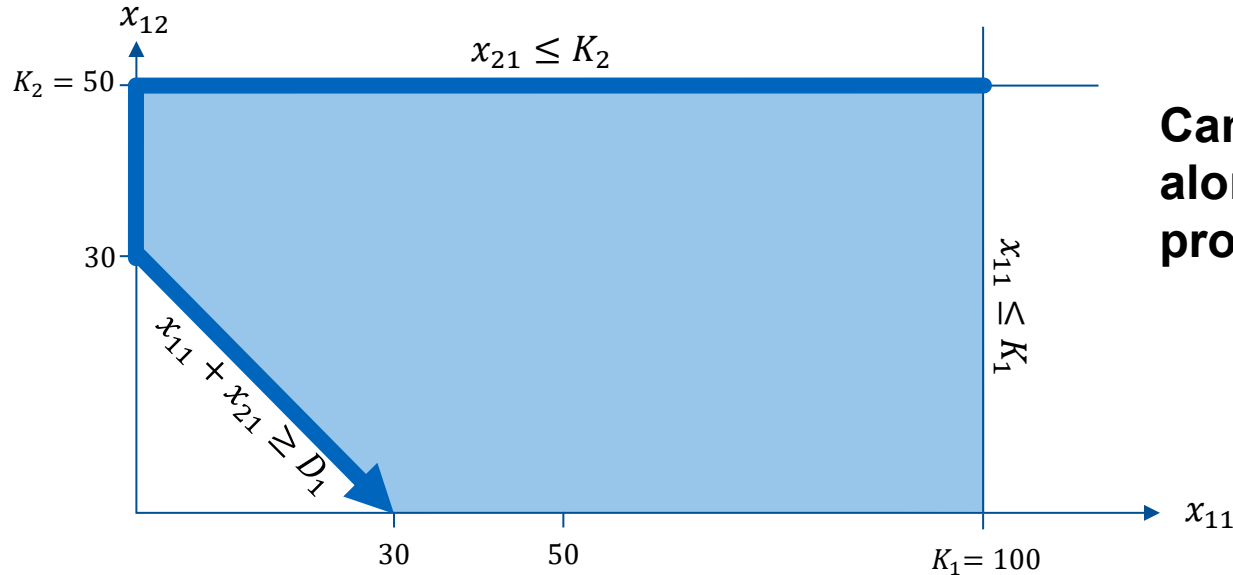
Number of variables (a)	Number of linear equations (b)	C_b^a
5	2	10
20	10	184,756
49	6	13,983,816
82	41	$4.25 \cdot 10^{23}$



Assume you can evaluate 1 million corners per second. Then this takes about 13.5 billion years. This is about the age of our universe.

Graphical Solution

$m = 2 \mid n = 1 \mid K_1 = 100, K_2 = 50 \mid D_1 = 30 \mid c_{11} = 80, c_{21} = 120$



**Can we find a fast way
along selected, most
promising corners?**

SC Network and Excel Solver

Small Example for SC Network Design Problem

$$\min \quad 25x_{11} + 30x_{21} + 35x_{12} + 40x_{22}$$

subject to:

$$x_{11} + x_{21} \geq 220$$

$$x_{12} + x_{22} \geq 180$$

$$x_{11} + x_{12} \leq 300$$

$$x_{21} + x_{22} \leq 500$$

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$$

Small Example for SC Network Design Problem

$$\min \quad 25x_{11} + 30x_{21} + 35x_{12} + 40x_{22}$$

subject to:

$$x_{11} + x_{21} \geq 220$$

$$x_{12} + x_{22} \geq 180$$

← Demand constraints

$$x_{11} + x_{12} \leq 300$$

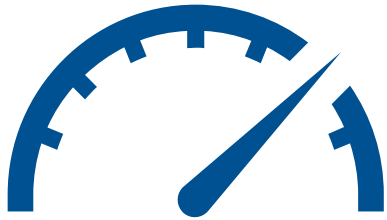
$$x_{21} + x_{22} \leq 500$$

← Capacity constraints

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$$

Computer Solvers

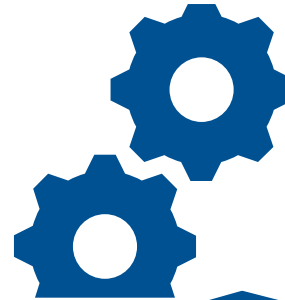
Why use solvers?



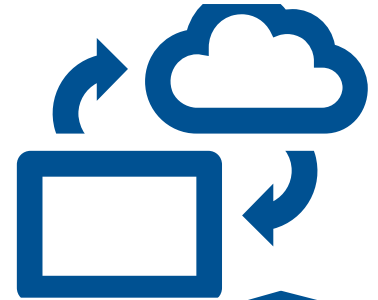
Fast



Reliable

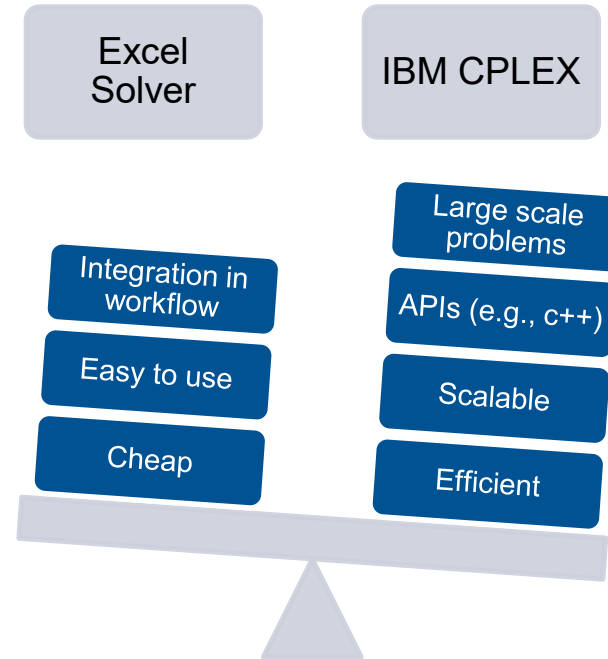


Automated



Scalable

Typical Solvers



Structure of Excel Worksheets before using Solver

Example for Excel Solver

$$\min \quad 25x_{11} + 30x_{21} + 35x_{12} + 40x_{22}$$

subject to:

$$x_{11} + x_{21} \geq 220$$

$$x_{12} + x_{22} \geq 180$$

$$x_{11} + x_{12} \leq 300$$

$$x_{21} + x_{22} \leq 500$$

$$x_{11}, x_{12}, x_{21}, x_{22} \geq 0$$

Insert Parameters

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1																
2																
3																
4																
5																
6				c _{ij}		Customer										
7						j=1		j=2								
8		Plant	i=1			25		35								
9			i=2			30		40								
10																
11						Customer location										
12				D _j		j=1		j=2								
13						220		180								
14																
15				K _i												
16		Plant	i=1			300										
17			i=2			400										
18																

Create Empty Cells for Decision Variables

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
19																
20			Decision variables													
21																
22						Customer										
23			x _{ij}		j=1		j=2									
24		Plant	i=1		0		0									
25			i=2		0		0									
26																
27																

Write up Objective Function as Formula

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1																
2																
3																
4																
5																
6																
7																
8																
9																
10																
11																
12																
13																
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29																
30																
31																
32																

Min z =

=SUMPRODUCT(E24:F25;E8:F9)

SUMPRODUCT(array1; [array2]; [array3]; [array4]; ...)

$$\sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij}$$

Write Constraints using Formulas for Terms

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
32											$\overleftarrow{i=1} \overleftarrow{j=1}$					
33																
34																
35																
36																
37																
38																
39																
40																
41																
42																
43																
44																

Constraints

Demand s

j=1 =SUM(E24:E25) >= 220

j=2 0 >= 180

$$\sum_{i=1}^m x_{ij} \geq D_j \quad j = 1, \dots, n$$

Capacity constrains

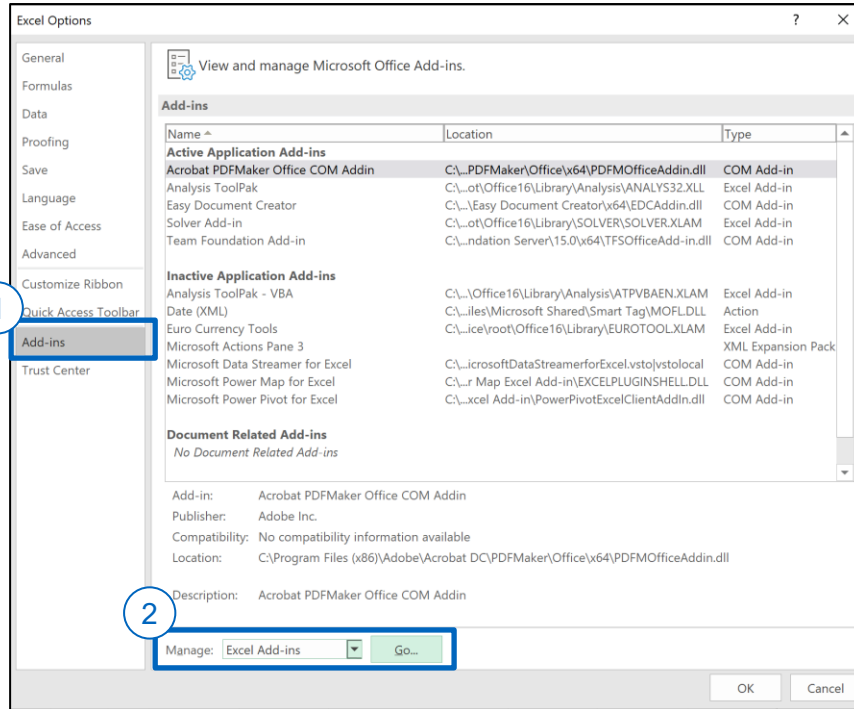
i=1 0 <= 300

i=2 0 <= 400

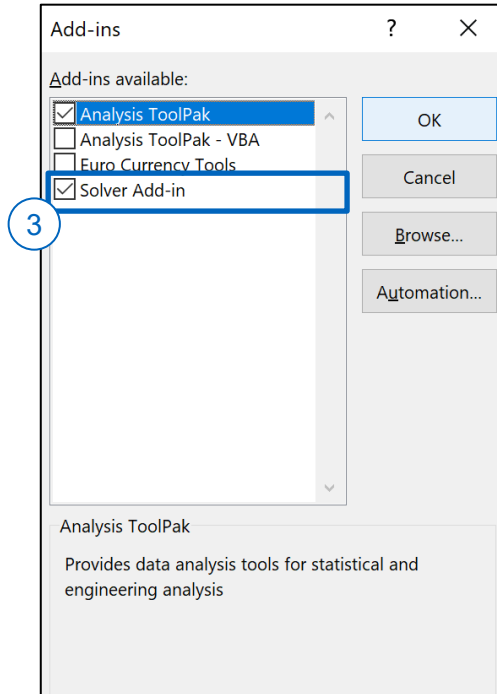
$$\sum_{j=1}^n x_{ij} \leq K_i \quad i = 1, \dots, m$$

Using Excel Solver

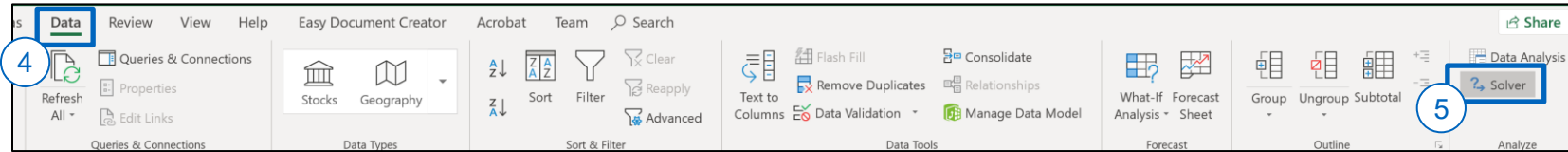
Activate Excel-Solver Plugin (File → Options)



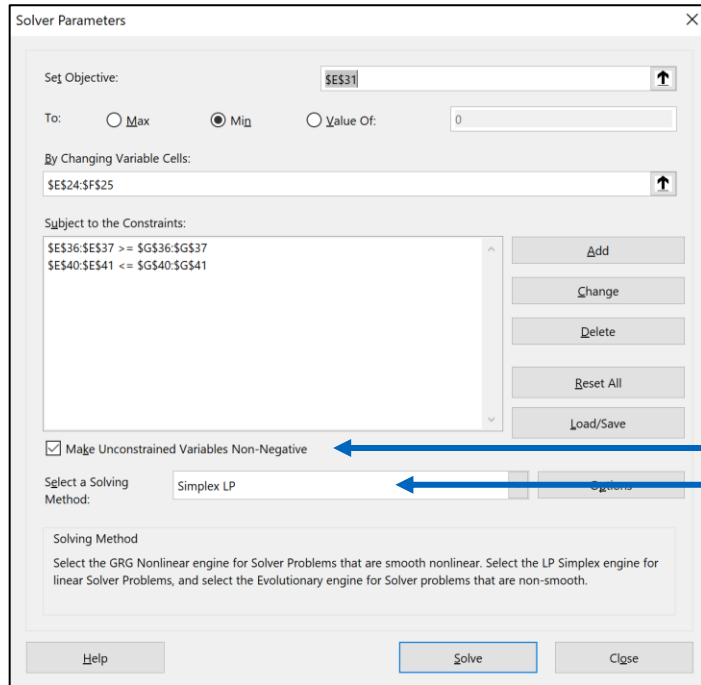
Activate Excel-Solver Plugin



Solver Appears in Data Tab



Tell Solver about your Model



The image shows the 'Solver Parameters' dialog box in Microsoft Excel. The 'Set Objective:' field contains '\$E\$31'. The 'To:' section has 'Min' selected. The 'By Changing Variable Cells:' field contains '\$E\$24:\$F\$25'. The 'Subject to the Constraints:' list contains two constraints: '\$E\$36:\$E\$37 >= \$G\$36:\$G\$37' and '\$E\$40:\$E\$41 <= \$G\$40:\$G\$41'. The 'Make Unconstrained Variables Non-Negative' checkbox is checked. The 'Select a Solving Method:' dropdown is set to 'Simplex LP'. The 'Solving Method' section provides instructions for selecting the GRG Nonlinear engine, LP Simplex engine, or Evolutionary engine. At the bottom are 'Help', 'Solve', and 'Close' buttons.

← Refer to the objective function which you entered as formula

← Refer to all decision variables

← Click on “Add” and follow instructions.

← Always keep activated

← Always use Simplex LP (most efficient method in Excel for linear problems)

Request a Sensitivity Report after Clicking “Solve”

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

☐ Return to Solver Parameters Dialog ☐ Outline Reports

Reports

- Answer
- Sensitivity**
- Limits

OK Cancel Save Scenario...

Reports

Creates the type of report that you specify, and places each report on a separate sheet in the workbook

← Perhaps the most useful report here

Solution

Objective function				
<u>Min z =</u>	12300			
Constraints				
<u>Demand satisfaction</u>				
j=1	220	>=	220	
j=2	180	>=	180	
<u>Capacity constraints</u>				
i=1	300	<=	300	
i=2	100	<=	400	

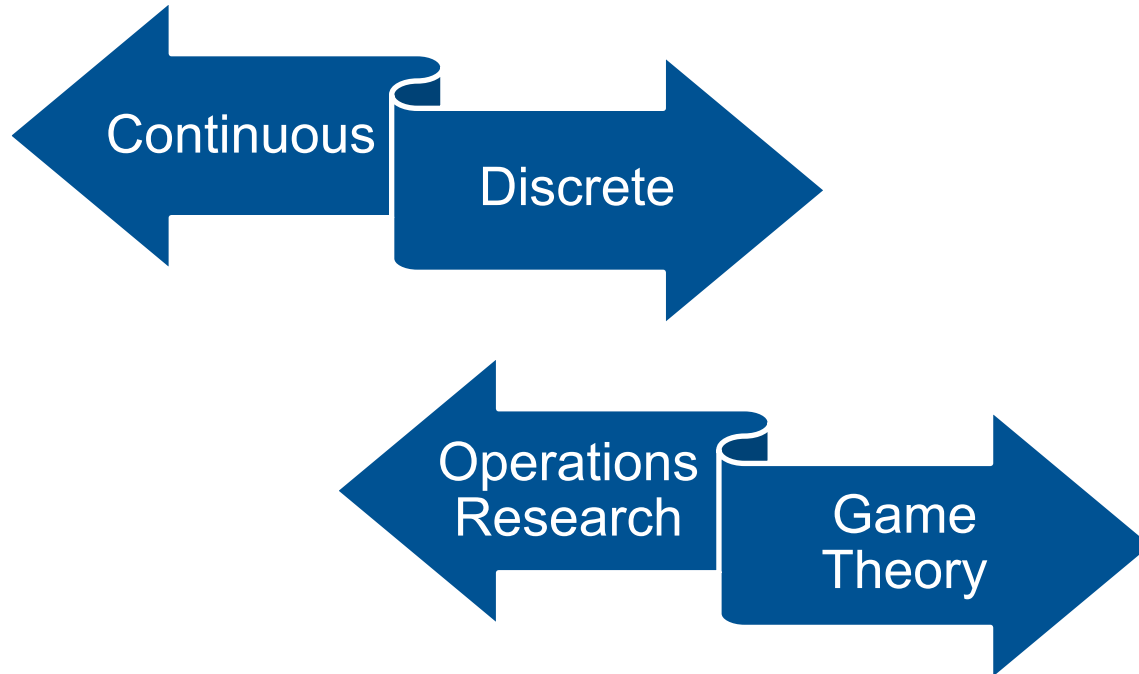
Sensitivity on New Spreadsheet

	A	B	C	D	E	F	G	H
1	Microsoft Excel 16.0 Sensitivity Report							
2	Worksheet: [Network_Problem_LP.xlsx]LP_small							
3	Report Created: 2019-06-07 16:42:03							
4								
5								
6	Variable Cells							
7			Final	Reduced	Objective	Allowable	Allowable	
8	Cell	Name	Value	Cost	Coefficient	Increase	Decrease	
9	\$E\$24	i=1 j=1	120	0	25	5	0	
10	\$F\$24	i=1 j=2	180	0	35	0	40	
11	\$E\$25	i=2 j=1	100	0	30	0	5	
12	\$F\$25	i=2 j=2	0	0	40	1E+30	0	
13								
14	Constraints							
15			Final	Shadow	Constraint	Allowable	Allowable	
16	Cell	Name	Value	Price	R.H. Side	Increase	Decrease	
17	\$E\$36	j=1 j=1	220	30	220	300	100	
18	\$E\$37	j=2 j=1	180	40	180	120	100	
19	\$E\$40	i=1 j=1	300	-5	300	100	120	
20	\$E\$41	i=2 j=1	100	0	400	1E+30	300	

We are usually most interested in the shadow price column.
Interpret these values!

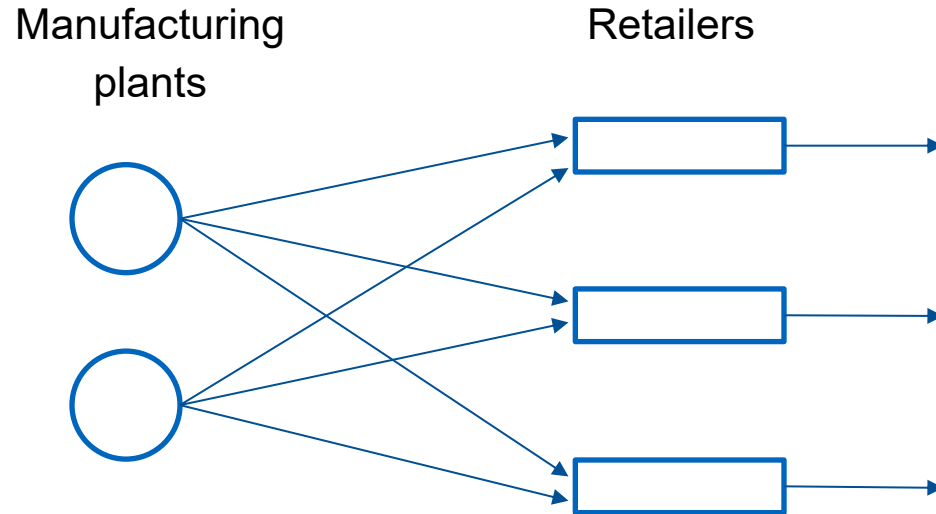
Facility Location Problems

Potential Facility Locations

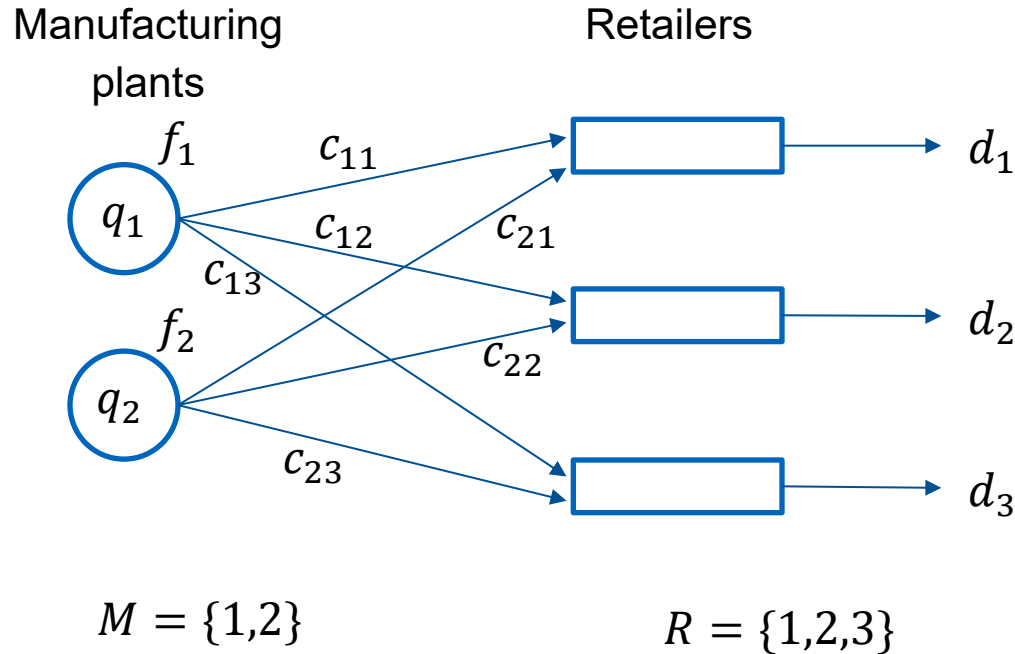


Single-Echelon Plant Location Model

Capacitated Plant Location Problem (CPLP)



Capacitated Plant Location Problem (CPLP)



Capacitated Plant Location Problem (CPLP)

$$\min \sum_{i \in M} \sum_{j \in R} c_{ij} x_{ij}$$

Amount shipped from
plant i to customer j



Capacitated Plant Location Problem (CPLP)

$$\min \sum_{i \in M} \sum_{j \in R} c_{ij} x_{ij} + \sum_{i \in M} f_i y_i$$

Amount shipped from plant i to customer j $\uparrow$ x_{ij}

Open up plant i ? $\uparrow$ y_i

Capacitated Plant Location Problem (CPLP)

$$\min \quad \sum_{i \in M} \sum_{j \in R} c_{ij} x_{ij} + \sum_{i \in M} f_i y_i$$

subject to:

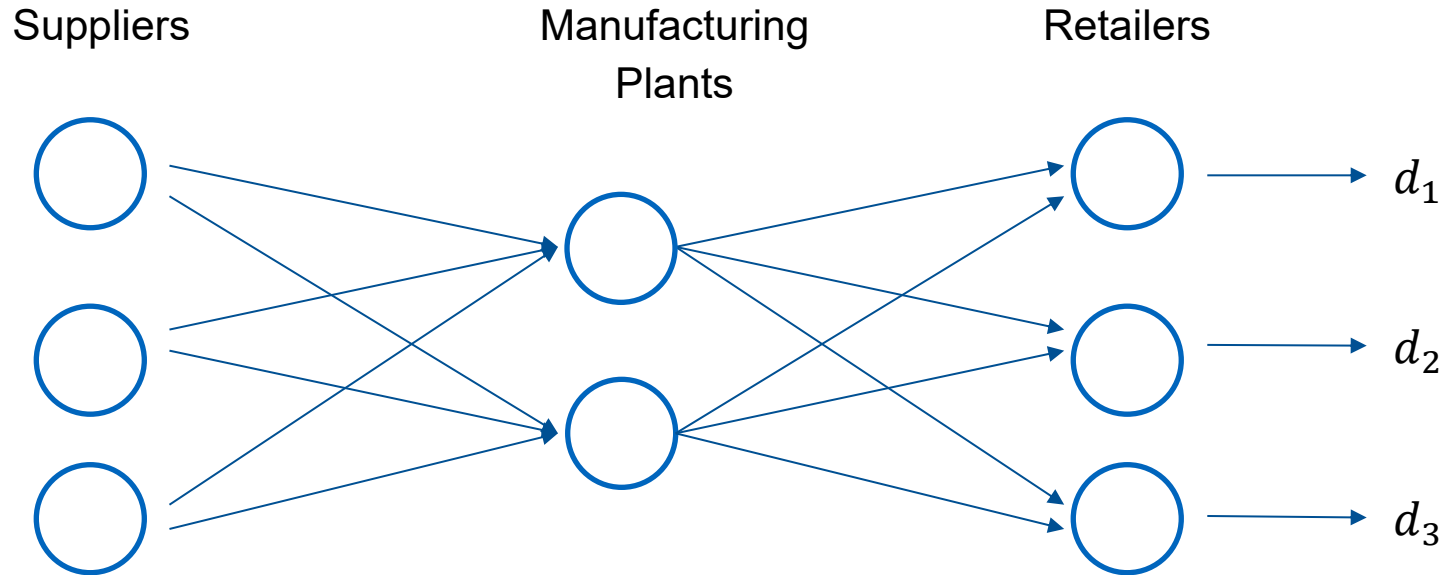
$$\sum_{i \in M} x_{ij} = d_j \quad , \quad j \in R$$

$$\sum_{j \in R} x_{ij} \leq q_i y_i \quad , \quad i \in M$$

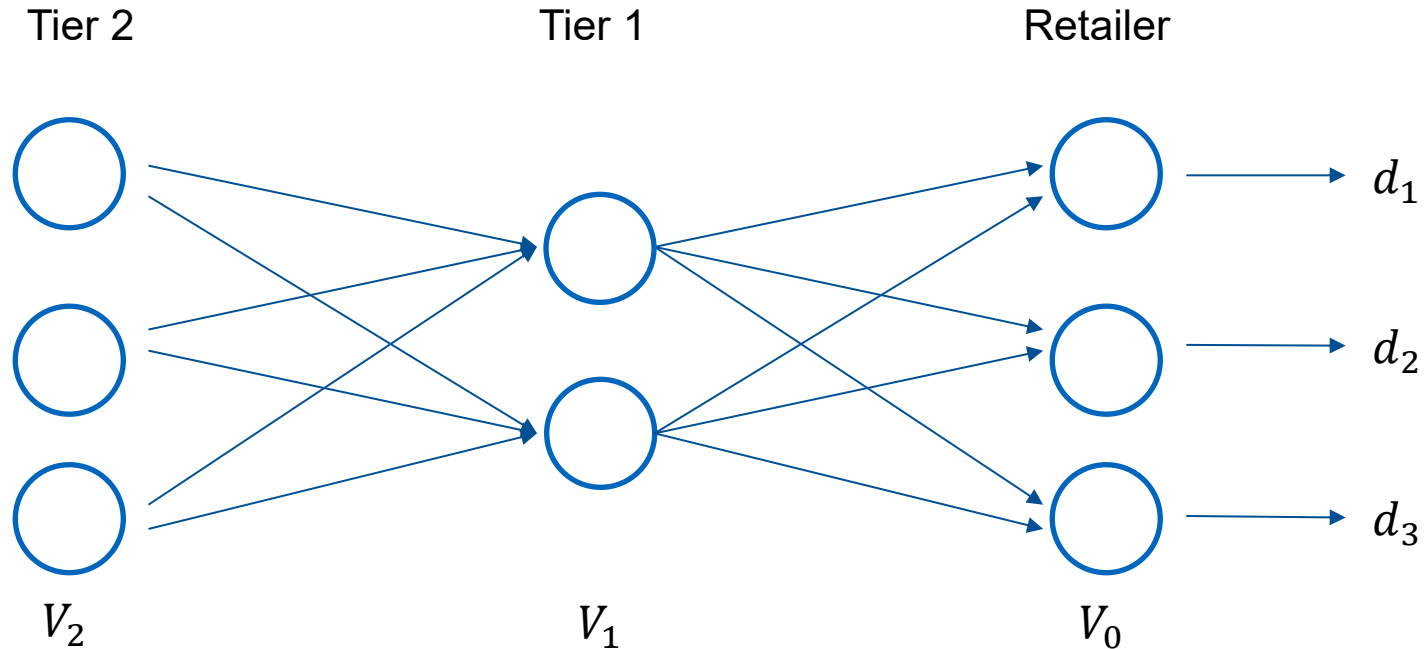
$$x_{ij} \geq 0,$$
$$y_i \in \{0,1\}$$

Two-Echelon Plant Location Model

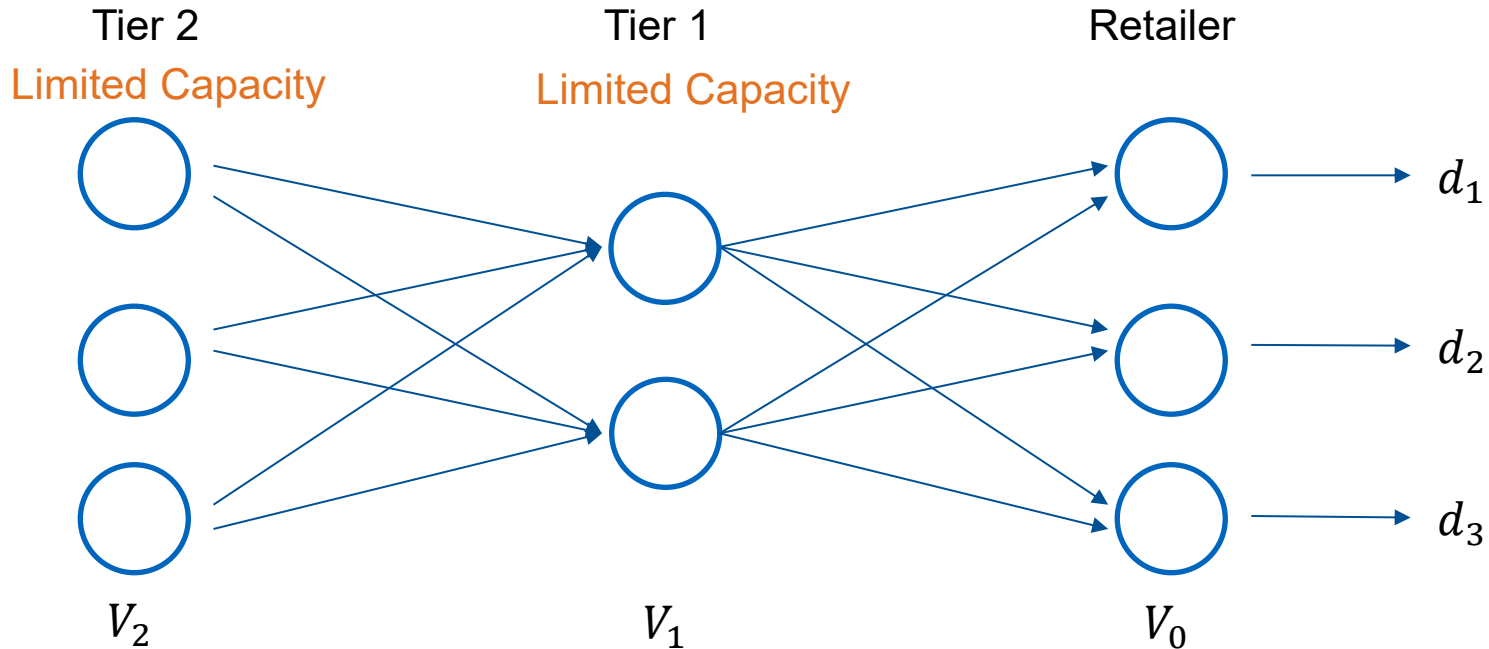
Two-echelon Plant Location Model



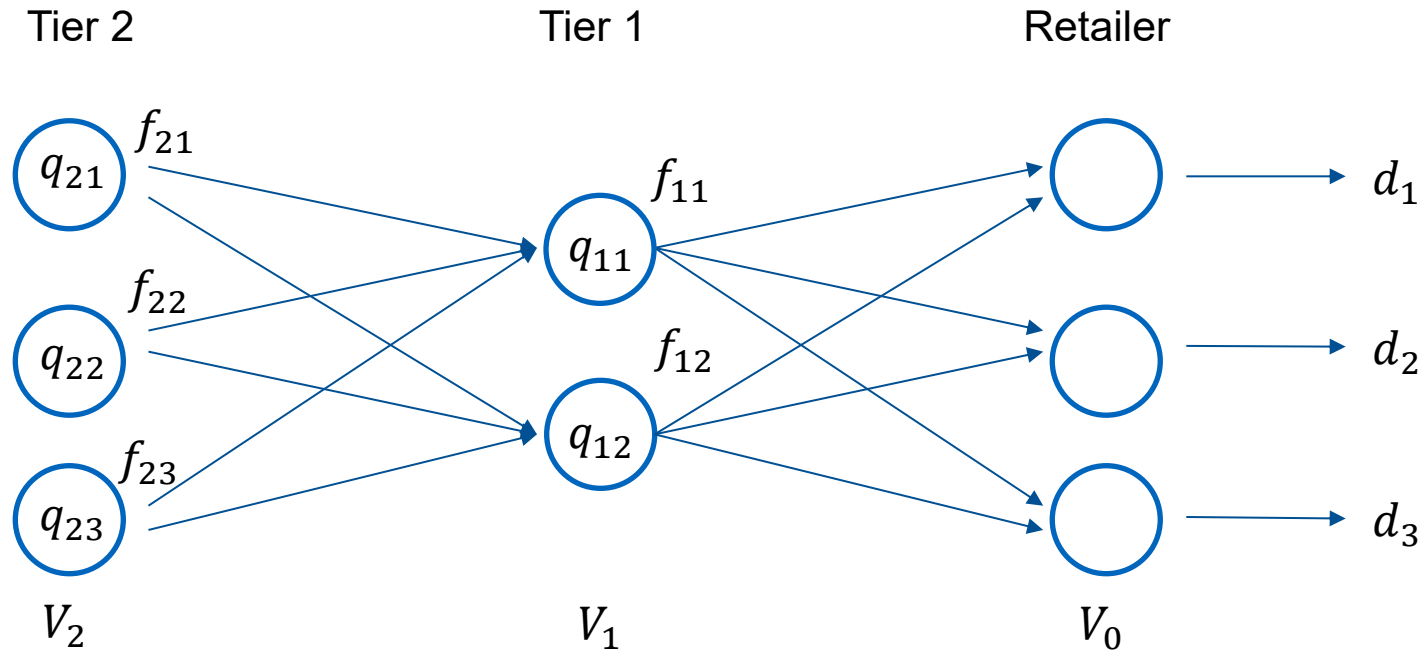
Two-Echelon Plant Location Model



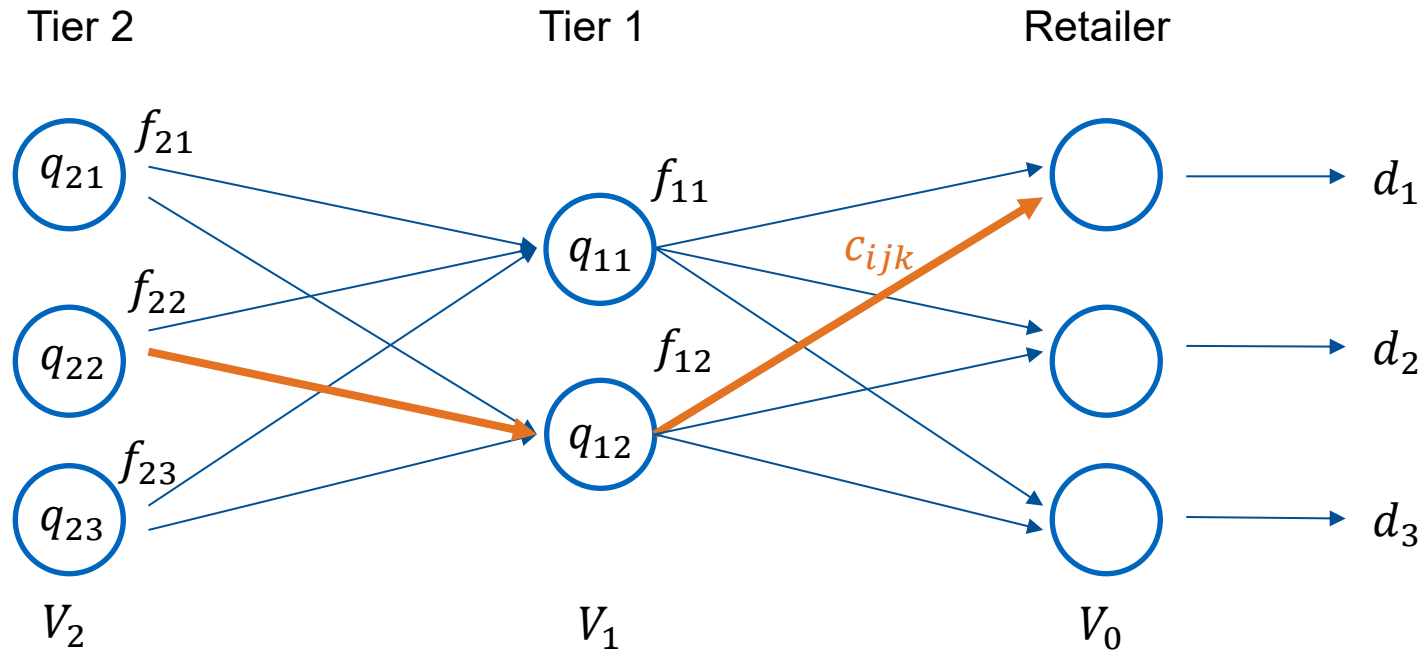
Two-Echelon Plant Location Model



Two-Echelon Plant Location Model



Two-Echelon Plant Location Model



Two-Echelon Plant Location Model

$$\min \sum_{i \in V_0} \sum_{j \in V_1} \sum_{k \in V_2} c_{ijk} x_{ijk} + \sum_{j \in V_1} f_{1j} y_{1j} + \sum_{k \in V_2} f_{2k} y_{2k}$$

subject to:

$$\sum_{j \in V_1} \sum_{k \in V_2} x_{ijk} = d_i \quad , \quad i \in V_0$$

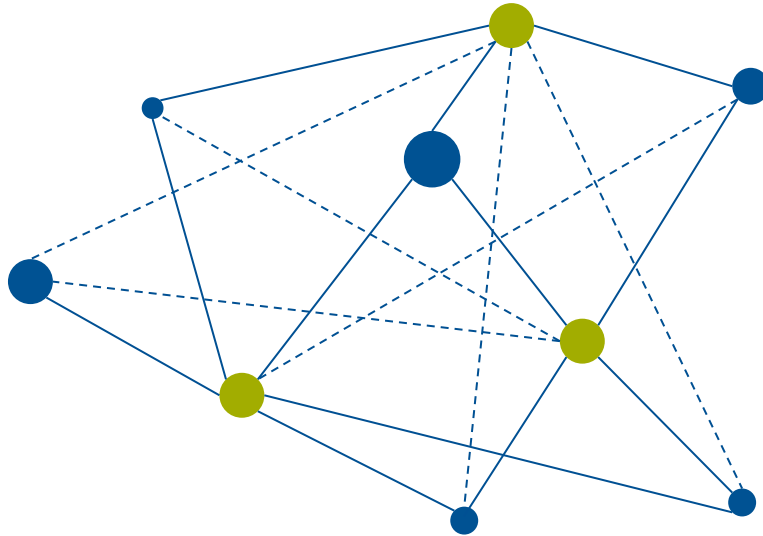
$$\sum_{i \in V_0} \sum_{k \in V_2} x_{ijk} \leq q_{1j} y_{1j} \quad , \quad j \in V_1$$

$$\sum_{i \in V_0} \sum_{j \in V_1} x_{ijk} \leq q_{2k} y_{2k} \quad , \quad k \in V_2$$

$$x_{ijk} \geq 0, \quad y_{1j} \in \{0,1\}, \quad y_{2k} \in \{0,1\}$$

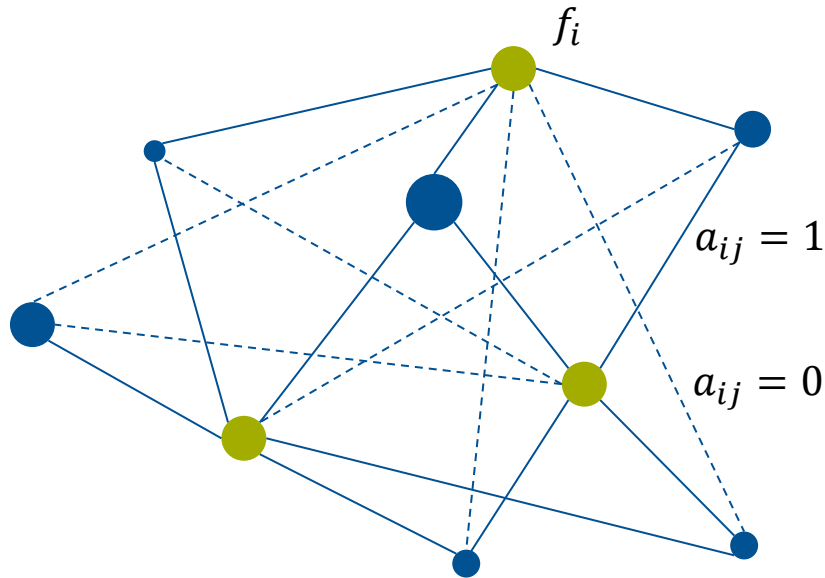
Location Covering Problems

Location Covering Problem



- Potential plant location
- Customer location
- How can you serve all customers?

Location Covering Problem



● Potential plant location

● Customer location

- How can you serve all customers?

f_i : Cost to open plant i

a_{ij} : can customer j be served from plant i ?

V_1 : Set of potential plants

V_2 : Set of customers

Location Covering Problem

$$\min \sum_{i \in V_1} f_i y_i$$

subject to:

$$\sum_{i \in V_1} a_{ij} y_i \geq 1 \quad , \quad j \in V_2$$

$$y_{ij} \in \{0,1\}$$

Location Covering Problem Heuristic

- Step 1:** For all plants with $f_i = 0$, set $y_i = 1$. Remove the constraints in which this plant appears.
- Step 2:** If plant i does not appear in any of the remaining constraints, set $y_i = 0$.
- Step 3:** If there are any constraints left, let n_i be the number of constraints in which plant i appears and calculate $\frac{f_i}{n_i}$ for each plant. Set $y_k = 1$ for the plant i for which $\frac{f_i}{n_i}$ is minimal. If there are multiple equal candidates, pick the one with the smallest value of i . Remove all constraints where plant k appears.
- Step 4:** If there are no more constraints, STOP, and set the remaining y_i to zero. Otherwise, go to step 2.

Location Covering Problem Heuristic EXAMPLE

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

$$y_1 + y_2 + y_3 + y_4 \geq 1$$

$$y_1 + y_3 + y_4 \geq 1$$

$$y_1 + y_2 + y_4 \geq 1$$

$$y_2 + y_4 \geq 1$$

$$y_1 + y_4 \geq 1$$

$$y_{ij} \in \{0,1\}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

Step 1

subject to:

$$y_1 + y_2 + y_3 + y_4 \geq 1$$

$$y_1 + y_3 + y_4 \geq 1$$

$$y_1 + y_2 + y_4 \geq 1$$

$$y_2 + y_4 \geq 1$$

$$y_1 + y_4 \geq 1$$

$$y_{ij} \in \{0,1\}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

Step 1

subject to:

$$y_3 = 1$$

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~

$$y_1 + y_2 + y_4 \geq 1$$

$$y_2 + y_4 \geq 1$$

$$y_1 + y_4 \geq 1$$

$$y_{ij} \in \{0,1\}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

Step 2

subject to:

$$y_3 = 1$$

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~

$$y_1 + y_2 + y_4 \geq 1$$

$$y_2 + y_4 \geq 1$$

$$y_1 + y_4 \geq 1$$

$$y_{ij} \in \{0,1\}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~

$$y_1 + y_2 + y_4 \geq 1$$

$$y_2 + y_4 \geq 1$$

$$y_1 + y_4 \geq 1$$

$$y_{ij} \in \{0,1\}$$

Step 3

$$y_3 = 1$$

$$\frac{f_1}{n_1} = \frac{2}{2} \quad \frac{f_2}{n_2} = \frac{3}{2} \quad \frac{f_4}{n_4} = \frac{4}{3}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

$$\begin{aligned} & \cancel{y_1 + y_2 + y_3 + y_4 \geq 1} \\ & \cancel{y_1 + y_3 + y_4 \geq 1} \\ & \underline{y_1 + y_2 + y_4 \geq 1} \\ & \quad y_2 + y_4 \geq 1 \\ & \underline{y_1 + y_4 \geq 1} \end{aligned}$$

$$y_{ij} \in \{0,1\}$$

Step 3

$$y_3 = 1$$

$$y_1 = 1$$

$$\boxed{\frac{f_1}{n_1} = \frac{2}{2}}$$

$$\frac{f_2}{n_2} = \frac{3}{2}$$

$$\frac{f_4}{n_4} = \frac{4}{3}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_2 + y_4 \geq 1$$~~

$$y_2 + y_4 \geq 1$$
~~$$y_1 + y_4 \geq 1$$~~

$$y_{ij} \in \{0,1\}$$

Step 4

$$y_3 = 1 \quad y_1 = 1$$

1 constraint left, hence go to step 2.

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

Step 2

$$y_3 = 1 \quad y_1 = 1$$

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_2 + y_4 \geq 1$$~~

$$y_2 + y_4 \geq 1$$
~~$$y_1 + y_4 \geq 1$$~~

$$y_{ij} \in \{0,1\}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_2 + y_4 \geq 1$$~~

$$y_2 + y_4 \geq 1$$
~~$$y_1 + y_4 \geq 1$$~~

$$y_{ij} \in \{0,1\}$$

Step 3

$$y_3 = 1 \quad y_1 = 1$$

$$\frac{f_2}{n_2} = \frac{3}{1} \quad \frac{f_4}{n_4} = \frac{4}{1}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_2 + y_4 \geq 1$$~~

$$y_2 + y_4 \geq 1$$
~~$$y_1 + y_4 \geq 1$$~~

$$y_{ij} \in \{0,1\}$$

Step 3

$$y_3 = 1 \quad y_1 = 1$$

$$\frac{f_2}{n_2} = \frac{3}{1} \quad \frac{f_4}{n_4} = \frac{4}{1}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

$$\begin{array}{l} \cancel{y_1 + y_2 + y_3 + y_4 \geq 1} \\ \cancel{y_1 + y_3 + y_4 \geq 1} \\ \cancel{y_1 + y_2 + y_4 \geq 1} \\ y_2 + y_4 \geq 1 \\ \cancel{y_1 + y_4 \geq 1} \end{array}$$

$$y_{ij} \in \{0,1\}$$

Step 3

$$y_3 = 1 \quad y_1 = 1 \quad y_2 = 1$$

$$\frac{f_2}{n_2} = \frac{3}{1} \quad \frac{f_4}{n_4} = \frac{4}{1}$$

Location Covering Problem Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

Step 4

$$y_3 = 1 \quad y_1 = 1 \quad y_2 = 1 \quad y_4 = 0$$

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_2 + y_4 \geq 1$$~~
~~$$y_2 + y_4 \geq 1$$~~
~~$$y_1 + y_4 \geq 1$$~~

$$y_{ij} \in \{0,1\}$$

Location Covering Problem: Heuristic **EXAMPLE**

$$\min \quad 2y_1 + 3y_2 + 0y_3 + 4y_4$$

subject to:

~~$$y_1 + y_2 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_3 + y_4 \geq 1$$~~
~~$$y_1 + y_2 + y_4 \geq 1$$~~
~~$$y_2 + y_4 \geq 1$$~~
~~$$y_1 + y_4 \geq 1$$~~

$$y_{ij} \in \{0,1\}$$

Solution

$$y_3 = 1 \quad y_1 = 1 \quad y_2 = 1 \quad y_4 = 0$$

Total cost

$$2 + 3 + 0 = 5$$

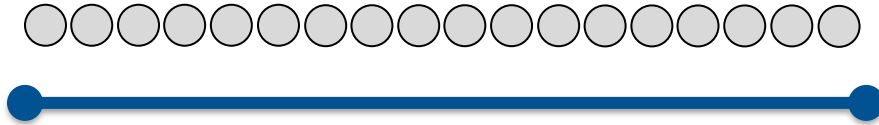
Optimal? – No!

Hotelling Competition

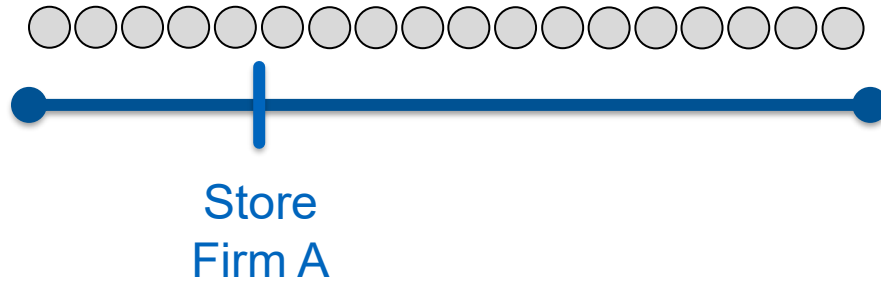
Hotelling Competition



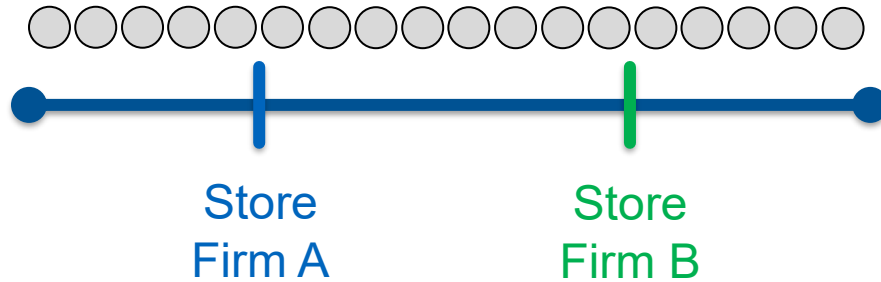
Hotelling Competition



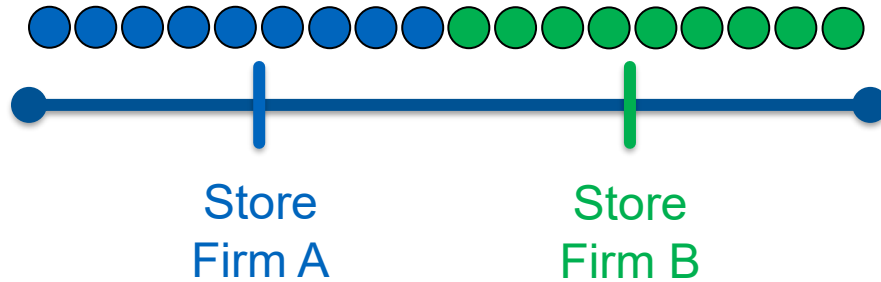
Hotelling Competition



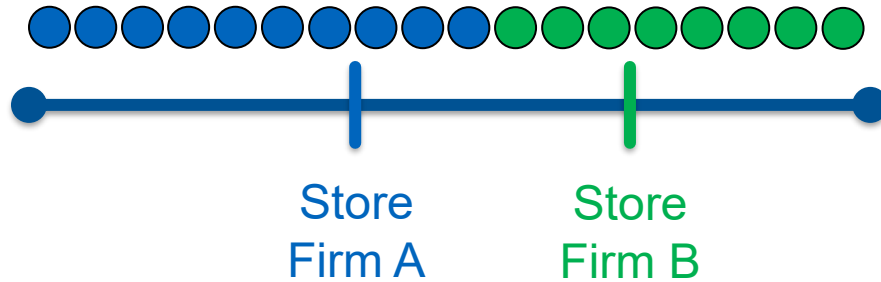
Hotelling Competition



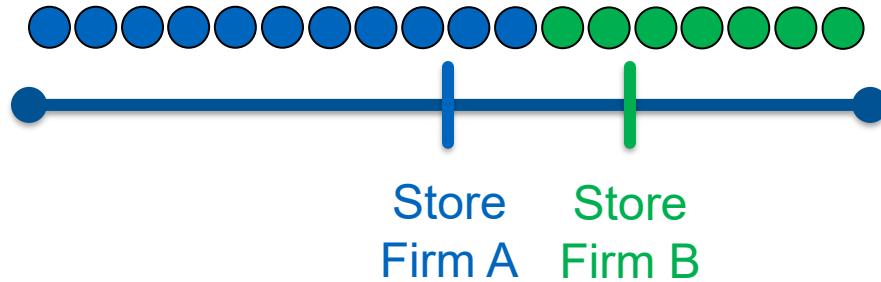
Hotelling Competition



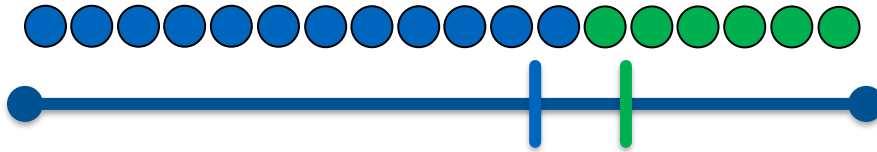
Hotelling Competition



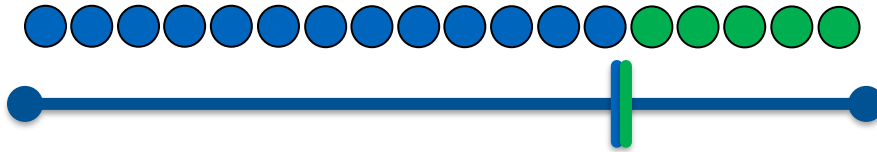
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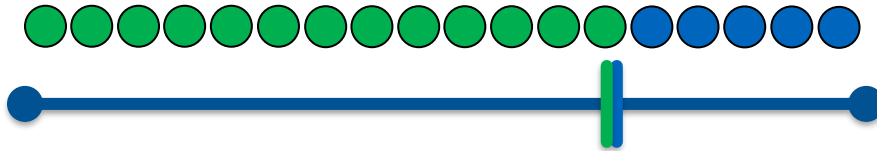
Hotelling Competition



Hotelling Competition



Hotelling Competition



Hotelling Competition

